

Theory of Algebraic Topology 13

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Lecture notes with worked examples in algebraic topology

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1 Problem 1 Products, Künneth Formula, Universal Coefficient Theorem, and Euler Characteristic

1.1 Problem 1: The space $X = L_4^3 \times \mathbb{RP}^3$

Problem. Let

$$X = L_4^3 \times \mathbb{RP}^3.$$

- Write out $C_*^{\text{cell}}(X)$ and use it to compute $H_*(X)$.
- Compute $H_*(X)$ using the Künneth formula and verify that it agrees with the answer in part (a).
- Use the universal coefficient theorem and the answer to part (b) to compute $H^*(X)$ and $H_*(X; \mathbb{Z}/2)$.
- Use the universal coefficient theorem to compute $H_*(L_4^3; \mathbb{Z}/2)$ and $H_*(\mathbb{RP}^3; \mathbb{Z}/2)$. Use this answer to compute $H_*(X; \mathbb{Z}/2)$ and verify that it agrees with the answer in part (c).

The cellular chain complexes of L_4^3 and $\mathbb{RP}^3$

The lens space L_4^3 has a CW-structure with one cell in each dimension

$$0, 1, 2, 3.$$

Its cellular chain complex is

$$0 \longrightarrow C_3(L_4^3) \cong \mathbb{Z} \xrightarrow{\partial_3=0} C_2(L_4^3) \cong \mathbb{Z} \xrightarrow{\partial_2=4} C_1(L_4^3) \cong \mathbb{Z} \xrightarrow{\partial_1=0} C_0(L_4^3) \cong \mathbb{Z} \longrightarrow 0.$$

Therefore

$$H_0(L_4^3) \cong \mathbb{Z}, \quad H_1(L_4^3) \cong \mathbb{Z}/4, \quad H_2(L_4^3) = 0, \quad H_3(L_4^3) \cong \mathbb{Z}.$$

Similarly, $\mathbb{RP}^3$ has a CW-structure with one cell in each dimension

$$0, 1, 2, 3.$$

Its cellular chain complex is

$$0 \longrightarrow C_3(\mathbb{RP}^3) \cong \mathbb{Z} \xrightarrow{\partial_3=0} C_2(\mathbb{RP}^3) \cong \mathbb{Z} \xrightarrow{\partial_2=2} C_1(\mathbb{RP}^3) \cong \mathbb{Z} \xrightarrow{\partial_1=0} C_0(\mathbb{RP}^3) \cong \mathbb{Z} \longrightarrow 0.$$

Hence

$$H_0(\mathbb{RP}^3) \cong \mathbb{Z}, \quad H_1(\mathbb{RP}^3) \cong \mathbb{Z}/2, \quad H_2(\mathbb{RP}^3) = 0, \quad H_3(\mathbb{RP}^3) \cong \mathbb{Z}.$$

Part (a): The cellular chain complex of X

The product CW-structure on

$$X = L_4^3 \times \mathbb{RP}^3$$

has one cell $e_{i,j}$ in dimension $i + j$ for each pair

$$0 \leq i \leq 3, \quad 0 \leq j \leq 3.$$

Here $e_{i,j}$ denotes the product of the i -cell of L_4^3 with the j -cell of $\mathbb{RP}^3$.

Thus

$$C_n^{\text{cell}}(X) \cong \bigoplus_{i+j=n} C_i^{\text{cell}}(L_4^3) \otimes C_j^{\text{cell}}(\mathbb{RP}^3).$$

Therefore the chain groups are

$$C_0(X) \cong \mathbb{Z},$$

$$C_1(X) \cong \mathbb{Z}^2,$$

$$C_2(X) \cong \mathbb{Z}^3,$$

$$C_3(X) \cong \mathbb{Z}^4,$$

$$C_4(X) \cong \mathbb{Z}^3,$$

$$C_5(X) \cong \mathbb{Z}^2,$$

$$C_6(X) \cong \mathbb{Z}.$$

The cellular differential on the product is given by

$$\partial(a \otimes b) = \partial a \otimes b + (-1)^{|a|} a \otimes \partial b.$$

We use the following ordered bases:

$$C_0 = \langle e_{0,0} \rangle,$$

$$C_1 = \langle e_{1,0}, e_{0,1} \rangle,$$

$$C_2 = \langle e_{2,0}, e_{1,1}, e_{0,2} \rangle,$$

$$C_3 = \langle e_{3,0}, e_{2,1}, e_{1,2}, e_{0,3} \rangle,$$

$$C_4 = \langle e_{3,1}, e_{2,2}, e_{1,3} \rangle,$$

$$C_5 = \langle e_{3,2}, e_{2,3} \rangle,$$

$$C_6 = \langle e_{3,3} \rangle.$$

With respect to these bases, the boundary maps are as follows.

First,

$$\partial_1 = 0.$$

Next,

$$\partial_2(e_{2,0}) = 4e_{1,0}, \quad \partial_2(e_{1,1}) = 0, \quad \partial_2(e_{0,2}) = 2e_{0,1}.$$

Hence

$$[\partial_2] = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

For ∂_3 , we get

$$\begin{aligned} \partial_3(e_{3,0}) &= 0, \\ \partial_3(e_{2,1}) &= 4e_{1,1}, \\ \partial_3(e_{1,2}) &= -2e_{1,1}, \\ \partial_3(e_{0,3}) &= 0. \end{aligned}$$

Thus

$$[\partial_3] = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 4 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

For ∂_4 , we have

$$\begin{aligned} \partial_4(e_{3,1}) &= 0, \\ \partial_4(e_{2,2}) &= 2e_{2,1} + 4e_{1,2}, \\ \partial_4(e_{1,3}) &= 0. \end{aligned}$$

So

$$[\partial_4] = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

For ∂_5 , we obtain

$$\partial_5(e_{3,2}) = -2e_{3,1}, \quad \partial_5(e_{2,3}) = 4e_{1,3}.$$

Hence

$$[\partial_5] = \begin{pmatrix} -2 & 0 \\ 0 & 0 \\ 0 & 4 \end{pmatrix}.$$

Finally,

$$\partial_6(e_{3,3}) = 0.$$

Therefore

$$\partial_6 = 0.$$

So the full cellular chain complex is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\partial_6=0} \mathbb{Z}^2 \xrightarrow{\partial_5} \mathbb{Z}^3 \xrightarrow{\partial_4} \mathbb{Z}^4 \xrightarrow{\partial_3} \mathbb{Z}^3 \xrightarrow{\partial_2} \mathbb{Z}^2 \xrightarrow{\partial_1=0} \mathbb{Z} \longrightarrow 0.$$

We now compute homology degree by degree.

In degree 0,

$$H_0(X) \cong \mathbb{Z}.$$

In degree 1,

$$H_1(X) = \ker \partial_1 / \operatorname{im} \partial_2 = \mathbb{Z}^2 / \langle 4e_{1,0}, 2e_{0,1} \rangle.$$

Hence

$$H_1(X) \cong \mathbb{Z}/4 \oplus \mathbb{Z}/2.$$

In degree 2, the map ∂_2 sends

$$ae_{2,0} + be_{1,1} + ce_{0,2}$$

to

$$4ae_{1,0} + 2ce_{0,1}.$$

Thus

$$\ker \partial_2 = \langle e_{1,1} \rangle \cong \mathbb{Z}.$$

Also,

$$\text{im } \partial_3 = \langle 4e_{1,1}, -2e_{1,1} \rangle = \langle 2e_{1,1} \rangle.$$

Therefore

$$H_2(X) \cong \mathbb{Z}/\langle 2 \rangle \cong \mathbb{Z}/2.$$

In degree 3, an element of C_3 has the form

$$ae_{3,0} + be_{2,1} + ce_{1,2} + de_{0,3}.$$

The condition that it lies in $\ker \partial_3$ is

$$4b - 2c = 0.$$

Equivalently,

$$c = 2b.$$

Thus

$$\ker \partial_3 = \langle e_{3,0}, e_{2,1} + 2e_{1,2}, e_{0,3} \rangle \cong \mathbb{Z}^3.$$

Meanwhile

$$\text{im } \partial_4 = \langle 2e_{2,1} + 4e_{1,2} \rangle = \langle 2(e_{2,1} + 2e_{1,2}) \rangle.$$

Therefore

$$H_3(X) \cong \mathbb{Z} \oplus \mathbb{Z}/2 \oplus \mathbb{Z} \cong \mathbb{Z}^2 \oplus \mathbb{Z}/2.$$

In degree 4, we have

$$\partial_4(ae_{3,1} + be_{2,2} + ce_{1,3}) = b(2e_{2,1} + 4e_{1,2}).$$

Therefore

$$\ker \partial_4 = \langle e_{3,1}, e_{1,3} \rangle \cong \mathbb{Z}^2.$$

Also,

$$\text{im } \partial_5 = \langle -2e_{3,1}, 4e_{1,3} \rangle.$$

Hence

$$H_4(X) \cong \mathbb{Z}/2 \oplus \mathbb{Z}/4.$$

Equivalently,

$$H_4(X) \cong \mathbb{Z}/4 \oplus \mathbb{Z}/2.$$

In degree 5,

$$\partial_5(ae_{3,2} + be_{2,3}) = -2ae_{3,1} + 4be_{1,3}.$$

This vanishes only if

$$a = 0, \quad b = 0.$$

Therefore

$$\ker \partial_5 = 0.$$

Since $\text{im } \partial_6 = 0$, we get

$$H_5(X) = 0.$$

Finally, in degree 6,

$$\partial_6 = 0,$$

and there are no 7-cells. Therefore

$$H_6(X) \cong \mathbb{Z}.$$

Thus

$$H_k(X) \cong \begin{cases} \mathbb{Z}, & k = 0, 6, \\ \mathbb{Z}/4 \oplus \mathbb{Z}/2, & k = 1, 4, \\ \mathbb{Z}/2, & k = 2, \\ \mathbb{Z}^2 \oplus \mathbb{Z}/2, & k = 3, \\ 0, & k = 5. \end{cases}$$

Part (b): Computation using the Künneth formula

The integral Künneth theorem gives a short exact sequence

$$0 \longrightarrow \bigoplus_{i+j=n} H_i(L_4^3) \otimes H_j(\mathbb{RP}^3) \longrightarrow H_n(X) \longrightarrow \bigoplus_{i+j=n-1} \text{Tor}(H_i(L_4^3), H_j(\mathbb{RP}^3)) \longrightarrow 0.$$

We use the standard facts

$$\mathbb{Z} \otimes G \cong G,$$

$$\mathbb{Z}/m \otimes \mathbb{Z}/n \cong \mathbb{Z}/\gcd(m, n),$$

and

$$\text{Tor}(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n).$$

Since

$$H_*(L_4^3) = (\mathbb{Z}, \mathbb{Z}/4, 0, \mathbb{Z}),$$

and

$$H_*(\mathbb{RP}^3) = (\mathbb{Z}, \mathbb{Z}/2, 0, \mathbb{Z}),$$

we compute degree by degree.

For $n = 0$,

$$H_0(X) \cong H_0(L_4^3) \otimes H_0(\mathbb{RP}^3) \cong \mathbb{Z}.$$

For $n = 1$,

$$H_1(X) \cong (H_1(L_4^3) \otimes H_0(\mathbb{RP}^3)) \oplus (H_0(L_4^3) \otimes H_1(\mathbb{RP}^3)).$$

Thus

$$H_1(X) \cong \mathbb{Z}/4 \oplus \mathbb{Z}/2.$$

For $n = 2$, the only nonzero tensor contribution is

$$H_1(L_4^3) \otimes H_1(\mathbb{RP}^3) \cong \mathbb{Z}/4 \otimes \mathbb{Z}/2 \cong \mathbb{Z}/2.$$

There is no nonzero Tor contribution in this degree. Therefore

$$H_2(X) \cong \mathbb{Z}/2.$$

For $n = 3$, the tensor contributions are

$$H_3(L_4^3) \otimes H_0(\mathbb{RP}^3) \cong \mathbb{Z},$$

and

$$H_0(L_4^3) \otimes H_3(\mathbb{RP}^3) \cong \mathbb{Z}.$$

The Tor contribution is

$$\text{Tor}(H_1(L_4^3), H_1(\mathbb{RP}^3)) = \text{Tor}(\mathbb{Z}/4, \mathbb{Z}/2) \cong \mathbb{Z}/2.$$

Hence

$$H_3(X) \cong \mathbb{Z}^2 \oplus \mathbb{Z}/2.$$

For $n = 4$, the tensor contributions are

$$H_3(L_4^3) \otimes H_1(\mathbb{RP}^3) \cong \mathbb{Z}/2,$$

and

$$H_1(L_4^3) \otimes H_3(\mathbb{RP}^3) \cong \mathbb{Z}/4.$$

There is no nonzero Tor contribution. Thus

$$H_4(X) \cong \mathbb{Z}/4 \oplus \mathbb{Z}/2.$$

For $n = 5$, all tensor and Tor terms vanish. Therefore

$$H_5(X) = 0.$$

For $n = 6$,

$$H_6(X) \cong H_3(L_4^3) \otimes H_3(\mathbb{RP}^3) \cong \mathbb{Z}.$$

Thus the Künneth formula gives

$$H_k(X) \cong \begin{cases} \mathbb{Z}, & k = 0, 6, \\ \mathbb{Z}/4 \oplus \mathbb{Z}/2, & k = 1, 4, \\ \mathbb{Z}/2, & k = 2, \\ \mathbb{Z}^2 \oplus \mathbb{Z}/2, & k = 3, \\ 0, & k = 5. \end{cases}$$

This agrees exactly with the computation using cellular chains.

Part (c): Integral cohomology and mod-2 homology of X

The universal coefficient theorem for cohomology says that there is a short exact sequence

$$0 \longrightarrow \text{Ext}(H_{n-1}(X), \mathbb{Z}) \longrightarrow H^n(X) \longrightarrow \text{Hom}(H_n(X), \mathbb{Z}) \longrightarrow 0.$$

We use

$$\text{Hom}(\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z}, \quad \text{Hom}(\mathbb{Z}/m, \mathbb{Z}) = 0,$$

and

$$\text{Ext}(\mathbb{Z}, \mathbb{Z}) = 0, \quad \text{Ext}(\mathbb{Z}/m, \mathbb{Z}) \cong \mathbb{Z}/m.$$

Since

$$\begin{aligned} H_0(X) &\cong \mathbb{Z}, \\ H_1(X) &\cong \mathbb{Z}/4 \oplus \mathbb{Z}/2, \\ H_2(X) &\cong \mathbb{Z}/2, \\ H_3(X) &\cong \mathbb{Z}^2 \oplus \mathbb{Z}/2, \\ H_4(X) &\cong \mathbb{Z}/4 \oplus \mathbb{Z}/2, \\ H_5(X) &= 0, \\ H_6(X) &\cong \mathbb{Z}, \end{aligned}$$

we obtain

$$\begin{aligned} H^0(X) &\cong \mathbb{Z}, \\ H^1(X) &= 0, \\ H^2(X) &\cong \mathbb{Z}/4 \oplus \mathbb{Z}/2, \\ H^3(X) &\cong \mathbb{Z}^2 \oplus \mathbb{Z}/2, \\ H^4(X) &\cong \mathbb{Z}/2, \\ H^5(X) &\cong \mathbb{Z}/4 \oplus \mathbb{Z}/2, \\ H^6(X) &\cong \mathbb{Z}. \end{aligned}$$

Therefore

$$H^k(X) \cong \begin{cases} \mathbb{Z}, & k = 0, 6, \\ 0, & k = 1, \\ \mathbb{Z}/4 \oplus \mathbb{Z}/2, & k = 2, 5, \\ \mathbb{Z}^2 \oplus \mathbb{Z}/2, & k = 3, \\ \mathbb{Z}/2, & k = 4. \end{cases}$$

Now we compute $H_*(X; \mathbb{Z}/2)$. The universal coefficient theorem for homology says that

$$0 \longrightarrow H_n(X) \otimes \mathbb{Z}/2 \longrightarrow H_n(X; \mathbb{Z}/2) \longrightarrow \text{Tor}(H_{n-1}(X), \mathbb{Z}/2) \longrightarrow 0.$$

Since the coefficient group is a field, the sequence splits, although not canonically. We use

$$\mathbb{Z} \otimes \mathbb{Z}/2 \cong \mathbb{Z}/2,$$

$$\mathbb{Z}/m \otimes \mathbb{Z}/2 \cong \mathbb{Z}/\gcd(m, 2),$$

and

$$\mathrm{Tor}(\mathbb{Z}/m, \mathbb{Z}/2) \cong \mathbb{Z}/\gcd(m, 2).$$

For $n = 0$,

$$H_0(X; \mathbb{Z}/2) \cong \mathbb{Z}/2.$$

For $n = 1$,

$$H_1(X) \otimes \mathbb{Z}/2 \cong (\mathbb{Z}/4 \oplus \mathbb{Z}/2) \otimes \mathbb{Z}/2 \cong (\mathbb{Z}/2)^2.$$

Also,

$$\mathrm{Tor}(H_0(X), \mathbb{Z}/2) = 0.$$

Hence

$$H_1(X; \mathbb{Z}/2) \cong (\mathbb{Z}/2)^2.$$

For $n = 2$,

$$H_2(X) \otimes \mathbb{Z}/2 \cong \mathbb{Z}/2,$$

and

$$\mathrm{Tor}(H_1(X), \mathbb{Z}/2) \cong \mathrm{Tor}(\mathbb{Z}/4, \mathbb{Z}/2) \oplus \mathrm{Tor}(\mathbb{Z}/2, \mathbb{Z}/2) \cong (\mathbb{Z}/2)^2.$$

Therefore

$$H_2(X; \mathbb{Z}/2) \cong (\mathbb{Z}/2)^3.$$

For $n = 3$,

$$H_3(X) \otimes \mathbb{Z}/2 \cong (\mathbb{Z}^2 \oplus \mathbb{Z}/2) \otimes \mathbb{Z}/2 \cong (\mathbb{Z}/2)^3.$$

Also,

$$\mathrm{Tor}(H_2(X), \mathbb{Z}/2) \cong \mathbb{Z}/2.$$

Thus

$$H_3(X; \mathbb{Z}/2) \cong (\mathbb{Z}/2)^4.$$

For $n = 4$,

$$H_4(X) \otimes \mathbb{Z}/2 \cong (\mathbb{Z}/4 \oplus \mathbb{Z}/2) \otimes \mathbb{Z}/2 \cong (\mathbb{Z}/2)^2.$$

Also,

$$\mathrm{Tor}(H_3(X), \mathbb{Z}/2) \cong \mathbb{Z}/2.$$

Therefore

$$H_4(X; \mathbb{Z}/2) \cong (\mathbb{Z}/2)^3.$$

For $n = 5$,

$$H_5(X) \otimes \mathbb{Z}/2 = 0,$$

and

$$\mathrm{Tor}(H_4(X), \mathbb{Z}/2) \cong \mathrm{Tor}(\mathbb{Z}/4, \mathbb{Z}/2) \oplus \mathrm{Tor}(\mathbb{Z}/2, \mathbb{Z}/2) \cong (\mathbb{Z}/2)^2.$$

Hence

$$H_5(X; \mathbb{Z}/2) \cong (\mathbb{Z}/2)^2.$$

For $n = 6$,

$$H_6(X) \otimes \mathbb{Z}/2 \cong \mathbb{Z}/2,$$

and

$$\mathrm{Tor}(H_5(X), \mathbb{Z}/2) = 0.$$

Thus

$$H_6(X; \mathbb{Z}/2) \cong \mathbb{Z}/2.$$

Therefore

$$H_k(X; \mathbb{Z}/2) \cong \begin{cases} \mathbb{Z}/2, & k = 0, 6, \\ (\mathbb{Z}/2)^2, & k = 1, 5, \\ (\mathbb{Z}/2)^3, & k = 2, 4, \\ (\mathbb{Z}/2)^4, & k = 3. \end{cases}$$

Part (d): Mod-2 homology of the factors and verification

First consider L_4^3 . We know

$$H_0(L_4^3) \cong \mathbb{Z}, \quad H_1(L_4^3) \cong \mathbb{Z}/4, \quad H_2(L_4^3) = 0, \quad H_3(L_4^3) \cong \mathbb{Z}.$$

Using the universal coefficient theorem for homology,

$$0 \longrightarrow H_n(L_4^3) \otimes \mathbb{Z}/2 \longrightarrow H_n(L_4^3; \mathbb{Z}/2) \longrightarrow \mathrm{Tor}(H_{n-1}(L_4^3), \mathbb{Z}/2) \longrightarrow 0,$$

we get

$$H_0(L_4^3; \mathbb{Z}/2) \cong \mathbb{Z}/2,$$

$$H_1(L_4^3; \mathbb{Z}/2) \cong \mathbb{Z}/2,$$

$$H_2(L_4^3; \mathbb{Z}/2) \cong \mathbb{Z}/2,$$

$$H_3(L_4^3; \mathbb{Z}/2) \cong \mathbb{Z}/2.$$

Thus

$$H_k(L_4^3; \mathbb{Z}/2) \cong \mathbb{Z}/2 \quad \text{for } k = 0, 1, 2, 3.$$

Now consider $\mathbb{RP}^3$. We know

$$H_0(\mathbb{RP}^3) \cong \mathbb{Z}, \quad H_1(\mathbb{RP}^3) \cong \mathbb{Z}/2, \quad H_2(\mathbb{RP}^3) = 0, \quad H_3(\mathbb{RP}^3) \cong \mathbb{Z}.$$

Again by the universal coefficient theorem,

$$H_0(\mathbb{RP}^3; \mathbb{Z}/2) \cong \mathbb{Z}/2,$$

$$H_1(\mathbb{RP}^3; \mathbb{Z}/2) \cong \mathbb{Z}/2,$$

$$H_2(\mathbb{RP}^3; \mathbb{Z}/2) \cong \mathbb{Z}/2,$$

$$H_3(\mathbb{RP}^3; \mathbb{Z}/2) \cong \mathbb{Z}/2.$$

Therefore

$$H_k(\mathbb{RP}^3; \mathbb{Z}/2) \cong \mathbb{Z}/2 \quad \text{for } k = 0, 1, 2, 3.$$

Now apply the Künneth formula over the field $\mathbb{Z}/2$. Since $\mathbb{Z}/2$ is a field, there are no Tor terms:

$$H_n(X; \mathbb{Z}/2) \cong \bigoplus_{i+j=n} H_i(L_4^3; \mathbb{Z}/2) \otimes_{\mathbb{Z}/2} H_j(\mathbb{RP}^3; \mathbb{Z}/2).$$

Each factor has one copy of $\mathbb{Z}/2$ in degrees 0, 1, 2, 3. Therefore the dimension of $H_n(X; \mathbb{Z}/2)$ is the number of pairs

$$(i, j)$$

such that

$$i + j = n, \quad 0 \leq i, j \leq 3.$$

These numbers are

$$1, 2, 3, 4, 3, 2, 1$$

in degrees

$$0, 1, 2, 3, 4, 5, 6.$$

Hence

$$H_k(X; \mathbb{Z}/2) \cong \begin{cases} \mathbb{Z}/2, & k = 0, 6, \\ (\mathbb{Z}/2)^2, & k = 1, 5, \\ (\mathbb{Z}/2)^3, & k = 2, 4, \\ (\mathbb{Z}/2)^4, & k = 3. \end{cases}$$

This agrees exactly with the computation in part (c).

2 Problem 2: Basic properties of the Euler characteristic

Problem. If X is a finite cell complex, define the Euler characteristic by

$$\chi(X) = \sum_k (-1)^k n_k,$$

where n_k is the number of k -cells in X . Prove the following properties of χ :

- (a) $\chi(X) = P_{\mathbb{F}}(X)|_{t=-1}$, where $\mathbb{F}$ is any field.
- (b) $\chi(X \times Y) = \chi(X)\chi(Y)$.
- (c) If A and B are subcomplexes of X , then

$$\chi(A \cup B) = \chi(A) + \chi(B) - \chi(A \cap B).$$

Part (a): Euler characteristic and the Poincaré polynomial

Let $\mathbb{F}$ be a field. The Poincaré polynomial of X over $\mathbb{F}$ is

$$P_{\mathbb{F}}(X) = \sum_k \beta_k^{\mathbb{F}}(X) t^k,$$

where

$$\beta_k^{\mathbb{F}}(X) = \dim_{\mathbb{F}} H_k(X; \mathbb{F})$$

is the k -th Betti number of X with coefficients in $\mathbb{F}$.

We want to prove that

$$\chi(X) = P_{\mathbb{F}}(X) \Big|_{t=-1}.$$

The cellular chain complex of X with coefficients in $\mathbb{F}$ is

$$\cdots \longrightarrow C_k(X; \mathbb{F}) \xrightarrow{\partial_k} C_{k-1}(X; \mathbb{F}) \longrightarrow \cdots.$$

Because X has n_k many k -cells, we have

$$C_k(X; \mathbb{F}) \cong \mathbb{F}^{n_k}.$$

Therefore

$$\dim_{\mathbb{F}} C_k(X; \mathbb{F}) = n_k.$$

For a finite-dimensional chain complex over a field, the alternating sum of the dimensions of the chain groups equals the alternating sum of the dimensions of the homology groups:

$$\sum_k (-1)^k \dim_{\mathbb{F}} C_k(X; \mathbb{F}) = \sum_k (-1)^k \dim_{\mathbb{F}} H_k(X; \mathbb{F}).$$

For completeness, we recall why this is true. Let

$$Z_k = \ker \partial_k$$

be the group of k -cycles, and let

$$B_k = \operatorname{im} \partial_{k+1}$$

be the group of k -boundaries. Then

$$H_k(X; \mathbb{F}) = Z_k / B_k.$$

Since all groups are finite-dimensional vector spaces over $\mathbb{F}$,

$$\dim Z_k = \dim H_k + \dim B_k.$$

Also, by rank-nullity applied to

$$\partial_k : C_k \rightarrow C_{k-1},$$

we have

$$\dim C_k = \dim Z_k + \dim B_{k-1}.$$

Therefore

$$\dim C_k = \dim H_k + \dim B_k + \dim B_{k-1}.$$

Taking alternating sums gives

$$\sum_k (-1)^k \dim C_k = \sum_k (-1)^k \dim H_k + \sum_k (-1)^k \dim B_k + \sum_k (-1)^k \dim B_{k-1}.$$

The last two sums cancel because

$$\sum_k (-1)^k \dim B_{k-1} = - \sum_k (-1)^k \dim B_k.$$

Hence

$$\sum_k (-1)^k \dim C_k = \sum_k (-1)^k \dim H_k.$$

Now

$$\chi(X) = \sum_k (-1)^k n_k = \sum_k (-1)^k \dim_{\mathbb{F}} C_k(X; \mathbb{F}).$$

Therefore

$$\chi(X) = \sum_k (-1)^k \dim_{\mathbb{F}} H_k(X; \mathbb{F}) = \sum_k (-1)^k \beta_k^{\mathbb{F}}(X).$$

But

$$P_{\mathbb{F}}(X) \Big|_{t=-1} = \sum_k \beta_k^{\mathbb{F}}(X) (-1)^k.$$

Hence

$$\boxed{\chi(X) = P_{\mathbb{F}}(X) \Big|_{t=-1}.$$

Part (b): Multiplicativity under products

We prove that

$$\chi(X \times Y) = \chi(X) \chi(Y).$$

Let X and Y be finite cell complexes. Suppose X has n_i cells of dimension i , and Y has m_j cells of dimension j .

The product CW-structure on $X \times Y$ has one $(i+j)$ -cell for each pair consisting of an i -cell of X and a j -cell of Y . Therefore, the number of k -cells of $X \times Y$ is

$$\sum_{i+j=k} n_i m_j.$$

Thus

$$\chi(X \times Y) = \sum_k (-1)^k \left(\sum_{i+j=k} n_i m_j \right).$$

Changing the order of summation gives

$$\chi(X \times Y) = \sum_{i,j} (-1)^{i+j} n_i m_j.$$

Since

$$(-1)^{i+j} = (-1)^i (-1)^j,$$

we get

$$\chi(X \times Y) = \left(\sum_i (-1)^i n_i \right) \left(\sum_j (-1)^j m_j \right).$$

Therefore

$$\chi(X \times Y) = \chi(X)\chi(Y).$$

Hence

$$\boxed{\chi(X \times Y) = \chi(X)\chi(Y).}$$

Equivalently, one can prove this using part (a) and the Künneth formula over a field. Indeed, over a field $\mathbb{F}$,

$$H_n(X \times Y; \mathbb{F}) \cong \bigoplus_{i+j=n} H_i(X; \mathbb{F}) \otimes_{\mathbb{F}} H_j(Y; \mathbb{F}).$$

Taking dimensions gives

$$\beta_n^{\mathbb{F}}(X \times Y) = \sum_{i+j=n} \beta_i^{\mathbb{F}}(X) \beta_j^{\mathbb{F}}(Y).$$

Hence

$$P_{\mathbb{F}}(X \times Y) = P_{\mathbb{F}}(X)P_{\mathbb{F}}(Y).$$

Evaluating at $t = -1$, we obtain

$$\chi(X \times Y) = P_{\mathbb{F}}(X \times Y)|_{t=-1} = P_{\mathbb{F}}(X)|_{t=-1} P_{\mathbb{F}}(Y)|_{t=-1}.$$

Using part (a), this becomes

$$\chi(X \times Y) = \chi(X)\chi(Y).$$

Part (c): Inclusion-exclusion for subcomplexes

Let A and B be subcomplexes of X . We want to prove that

$$\chi(A \cup B) = \chi(A) + \chi(B) - \chi(A \cap B).$$

Because A and B are subcomplexes, their union $A \cup B$ and their intersection $A \cap B$ are also cell complexes built from cells of X .

For each k , let

$$n_k(A)$$

denote the number of k -cells of A , and similarly define

$$n_k(B), \quad n_k(A \cup B), \quad n_k(A \cap B).$$

For each fixed dimension k , the ordinary inclusion-exclusion principle for finite sets gives

$$n_k(A \cup B) = n_k(A) + n_k(B) - n_k(A \cap B).$$

Now take alternating sums over all k :

$$\chi(A \cup B) = \sum_k (-1)^k n_k(A \cup B).$$

Substituting the formula above gives

$$\chi(A \cup B) = \sum_k (-1)^k (n_k(A) + n_k(B) - n_k(A \cap B)).$$

Therefore

$$\chi(A \cup B) = \sum_k (-1)^k n_k(A) + \sum_k (-1)^k n_k(B) - \sum_k (-1)^k n_k(A \cap B).$$

Hence

$$\chi(A \cup B) = \chi(A) + \chi(B) - \chi(A \cap B).$$

Thus

$$\boxed{\chi(A \cup B) = \chi(A) + \chi(B) - \chi(A \cap B).}$$

Summary

For

$$X = L_4^3 \times \mathbb{RP}^3,$$

we have

$$H_k(X) \cong \begin{cases} \mathbb{Z}, & k = 0, 6, \\ \mathbb{Z}/4 \oplus \mathbb{Z}/2, & k = 1, 4, \\ \mathbb{Z}/2, & k = 2, \\ \mathbb{Z}^2 \oplus \mathbb{Z}/2, & k = 3, \\ 0, & k = 5. \end{cases}$$

Also,

$$H^k(X) \cong \begin{cases} \mathbb{Z}, & k = 0, 6, \\ 0, & k = 1, \\ \mathbb{Z}/4 \oplus \mathbb{Z}/2, & k = 2, 5, \\ \mathbb{Z}^2 \oplus \mathbb{Z}/2, & k = 3, \\ \mathbb{Z}/2, & k = 4. \end{cases}$$

Finally,

$$H_k(X; \mathbb{Z}/2) \cong \begin{cases} \mathbb{Z}/2, & k = 0, 6, \\ (\mathbb{Z}/2)^2, & k = 1, 5, \\ (\mathbb{Z}/2)^3, & k = 2, 4, \\ (\mathbb{Z}/2)^4, & k = 3. \end{cases}$$

For every finite cell complex X ,

$$\chi(X) = P_{\mathbb{F}}(X) \Big|_{t=-1}$$

for any field $\mathbb{F}$. Moreover,

$$\chi(X \times Y) = \chi(X)\chi(Y)$$

and, for subcomplexes $A, B \subseteq X$,

$$\chi(A \cup B) = \chi(A) + \chi(B) - \chi(A \cap B).$$

3 Self-contained theory for products, Künneth formula, UCT, and Euler characteristic

3.1 Cell complexes and cellular chains

A **CW complex** is a topological space built by attaching cells dimension by dimension. A k -cell is a copy of the open ball

$$e^k \cong \mathring{D}^k.$$

The closed disk D^k is attached along its boundary sphere

$$\partial D^k = S^{k-1}.$$

Thus a CW complex is constructed in stages:

$$X^0 \subseteq X^1 \subseteq X^2 \subseteq \dots,$$

where X^k is called the k -skeleton of X . The space X^k is obtained from X^{k-1} by attaching k -cells.

The cellular chain group in dimension k is the free abelian group generated by the k -cells:

$$C_k^{\text{cell}}(X) \cong \mathbb{Z}^{n_k},$$

where n_k is the number of k -cells of X .

The cellular chain complex is

$$\dots \longrightarrow C_k^{\text{cell}}(X) \xrightarrow{\partial_k} C_{k-1}^{\text{cell}}(X) \longrightarrow \dots \longrightarrow C_1^{\text{cell}}(X) \xrightarrow{\partial_1} C_0^{\text{cell}}(X) \longrightarrow 0.$$

The homology groups are

$$H_k(X) = \ker \partial_k / \text{im } \partial_{k+1}.$$

Here

$$Z_k(X) = \ker \partial_k$$

is the group of k -cycles, and

$$B_k(X) = \text{im } \partial_{k+1}$$

is the group of k -boundaries.

Thus

$$H_k(X) = Z_k(X)/B_k(X).$$

Intuitively:

- k -cycles are closed k -dimensional objects;
- k -boundaries are cycles that bound $(k+1)$ -dimensional objects;
- homology measures cycles modulo those cycles that are already boundaries.

3.2 Cellular chain complex of $\mathbb{RP}^n$

Real projective space is defined as

$$\mathbb{RP}^n = S^n/(x \sim -x).$$

It identifies antipodal points on the sphere.

There is a standard CW decomposition of $\mathbb{RP}^n$ with one cell in every dimension

$$0, 1, \dots, n.$$

Hence

$$C_k^{\text{cell}}(\mathbb{RP}^n) \cong \mathbb{Z} \quad \text{for } 0 \leq k \leq n.$$

The cellular boundary maps alternate between multiplication by 2 and multiplication by 0:

$$\partial_k = \begin{cases} 0, & k \text{ odd,} \\ 2, & k \text{ even.} \end{cases}$$

For $\mathbb{RP}^3$, the cellular chain complex is therefore

$$0 \longrightarrow \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0.$$

So

$$C_3(\mathbb{RP}^3) \cong \mathbb{Z}, \quad C_2(\mathbb{RP}^3) \cong \mathbb{Z}, \quad C_1(\mathbb{RP}^3) \cong \mathbb{Z}, \quad C_0(\mathbb{RP}^3) \cong \mathbb{Z}.$$

The homology groups are

$$\begin{aligned} H_0(\mathbb{RP}^3) &\cong \mathbb{Z}, \\ H_1(\mathbb{RP}^3) &\cong \mathbb{Z}/2, \\ H_2(\mathbb{RP}^3) &= 0, \\ H_3(\mathbb{RP}^3) &\cong \mathbb{Z}. \end{aligned}$$

The group $H_1(\mathbb{RP}^3) \cong \mathbb{Z}/2$ records the fact that the fundamental one-dimensional loop in $\mathbb{RP}^3$ has order 2. Going around twice becomes null-homologous.

3.3 Cellular chain complex of the lens space L_p^3

A lens space L_p^3 is a closed 3-manifold whose first homology is

$$H_1(L_p^3) \cong \mathbb{Z}/p.$$

For the lens space L_p^3 , there is a CW decomposition with one cell in every dimension

$$0, 1, 2, 3.$$

The cellular chain complex has the form

$$0 \longrightarrow \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{p} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0.$$

For L_4^3 , this becomes

$$0 \longrightarrow \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{4} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0.$$

Therefore

$$\begin{aligned} H_0(L_4^3) &\cong \mathbb{Z}, \\ H_1(L_4^3) &\cong \mathbb{Z}/4, \\ H_2(L_4^3) &= 0, \\ H_3(L_4^3) &\cong \mathbb{Z}. \end{aligned}$$

The torsion group

$$H_1(L_4^3) \cong \mathbb{Z}/4$$

means that there is a one-dimensional homology class whose fourth multiple is zero.

3.4 Product CW complexes

If X and Y are CW complexes, then $X \times Y$ has a natural product CW structure.

If e^i is an i -cell of X , and f^j is a j -cell of Y , then

$$e^i \times f^j$$

is an $(i + j)$ -cell of $X \times Y$.

Therefore,

$$C_n^{\text{cell}}(X \times Y) \cong \bigoplus_{i+j=n} C_i^{\text{cell}}(X) \otimes C_j^{\text{cell}}(Y).$$

The boundary map on the tensor product chain complex is

$$\partial(a \otimes b) = \partial a \otimes b + (-1)^{|a|} a \otimes \partial b,$$

where $|a|$ denotes the degree of a .

The sign

$$(-1)^{|a|}$$

is essential. It ensures that

$$\partial^2 = 0.$$

For

$$X = L_4^3 \times \mathbb{RP}^3,$$

each factor has one cell in dimensions 0, 1, 2, 3. Thus the number of cells in each dimension of the product is given by the number of pairs (i, j) with

$$i + j = n, \quad 0 \leq i, j \leq 3.$$

Hence the cellular chain groups have ranks

$$1, 2, 3, 4, 3, 2, 1$$

in dimensions

$$0, 1, 2, 3, 4, 5, 6.$$

So

$$C_0(X) \cong \mathbb{Z},$$

$$C_1(X) \cong \mathbb{Z}^2,$$

$$C_2(X) \cong \mathbb{Z}^3,$$

$$C_3(X) \cong \mathbb{Z}^4,$$

$$C_4(X) \cong \mathbb{Z}^3,$$

$$C_5(X) \cong \mathbb{Z}^2,$$

$$C_6(X) \cong \mathbb{Z}.$$

This gives a cellular chain complex

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}^2 \longrightarrow \mathbb{Z}^3 \longrightarrow \mathbb{Z}^4 \longrightarrow \mathbb{Z}^3 \longrightarrow \mathbb{Z}^2 \longrightarrow \mathbb{Z} \longrightarrow 0.$$

The actual maps depend on the cellular boundary maps of the two factors.

3.5 Tensor products of cyclic groups

The Künneth formula uses tensor products of abelian groups. The most important rules in this problem are:

$$\mathbb{Z} \otimes A \cong A,$$

$$A \otimes \mathbb{Z} \cong A,$$

and

$$\mathbb{Z}/m \otimes \mathbb{Z}/n \cong \mathbb{Z}/\gcd(m, n).$$

For example,

$$\mathbb{Z}/4 \otimes \mathbb{Z}/2 \cong \mathbb{Z}/2.$$

Also,

$$\mathbb{Z}/4 \otimes \mathbb{Z} \cong \mathbb{Z}/4,$$

and

$$\mathbb{Z}/2 \otimes \mathbb{Z} \cong \mathbb{Z}/2.$$

If A is free abelian, then tensoring with A does not create torsion beyond what is already present. For instance,

$$\mathbb{Z}^r \otimes G \cong G^r.$$

3.6 The Tor functor for cyclic groups

The integral Künneth formula also involves the functor

$$\mathrm{Tor}(-, -).$$

For cyclic groups,

$$\mathrm{Tor}(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n).$$

In particular,

$$\mathrm{Tor}(\mathbb{Z}/4, \mathbb{Z}/2) \cong \mathbb{Z}/2.$$

Also,

$$\mathrm{Tor}(\mathbb{Z}, A) = 0$$

for every abelian group A , because $\mathbb{Z}$ is free.

Therefore,

$$\mathrm{Tor}(\mathbb{Z}, \mathbb{Z}/2) = 0,$$

$$\mathrm{Tor}(\mathbb{Z}, \mathbb{Z}/4) = 0.$$

The Tor term measures hidden torsion interaction between the homology groups of two spaces.

For example, in the product

$$L_4^3 \times \mathbb{RP}^3,$$

the torsion groups

$$H_1(L_4^3) \cong \mathbb{Z}/4$$

and

$$H_1(\mathbb{RP}^3) \cong \mathbb{Z}/2$$

interact to produce

$$\mathrm{Tor}(\mathbb{Z}/4, \mathbb{Z}/2) \cong \mathbb{Z}/2.$$

This is exactly the extra $\mathbb{Z}/2$ contribution that appears in degree 3 of the product.

3.7 The integral Künneth formula

Let X and Y be spaces whose homology groups are finitely generated. The integral Künneth formula gives a short exact sequence

$$0 \longrightarrow \bigoplus_{i+j=n} H_i(X) \otimes H_j(Y) \longrightarrow H_n(X \times Y) \longrightarrow \bigoplus_{i+j=n-1} \text{Tor}(H_i(X), H_j(Y)) \longrightarrow 0.$$

This sequence splits, but not naturally. Therefore, as a group,

$$H_n(X \times Y) \cong \left(\bigoplus_{i+j=n} H_i(X) \otimes H_j(Y) \right) \oplus \left(\bigoplus_{i+j=n-1} \text{Tor}(H_i(X), H_j(Y)) \right),$$

although the splitting is not canonical.

The formula says that homology of a product has two types of contributions:

1. Tensor contributions

$$H_i(X) \otimes H_j(Y), \quad i + j = n.$$

These come from taking an i -dimensional class in X and a j -dimensional class in Y , producing an $(i + j)$ -dimensional class in $X \times Y$.

2. Tor contributions

$$\text{Tor}(H_i(X), H_j(Y)), \quad i + j = n - 1.$$

These arise from torsion interactions between the homology of X and the homology of Y .

For the space

$$X = L_4^3 \times \mathbb{RP}^3,$$

the nonzero integral homology groups of the factors are

$$H_0(L_4^3) \cong \mathbb{Z}, \quad H_1(L_4^3) \cong \mathbb{Z}/4, \quad H_3(L_4^3) \cong \mathbb{Z},$$

and

$$H_0(\mathbb{RP}^3) \cong \mathbb{Z}, \quad H_1(\mathbb{RP}^3) \cong \mathbb{Z}/2, \quad H_3(\mathbb{RP}^3) \cong \mathbb{Z}.$$

The most important torsion interaction is

$$\text{Tor}(H_1(L_4^3), H_1(\mathbb{RP}^3)) = \text{Tor}(\mathbb{Z}/4, \mathbb{Z}/2) \cong \mathbb{Z}/2.$$

3.8 The Künneth formula over a field

If $\mathbb{F}$ is a field, then the Künneth formula becomes simpler:

$$H_n(X \times Y; \mathbb{F}) \cong \bigoplus_{i+j=n} H_i(X; \mathbb{F}) \otimes_{\mathbb{F}} H_j(Y; \mathbb{F}).$$

There is no Tor term over a field.

This is because homology groups over a field are vector spaces, and vector spaces are free modules over the field.

Therefore, if

$$\dim_{\mathbb{F}} H_i(X; \mathbb{F}) = \beta_i(X; \mathbb{F}),$$

and

$$\dim_{\mathbb{F}} H_j(Y; \mathbb{F}) = \beta_j(Y; \mathbb{F}),$$

then

$$\dim_{\mathbb{F}} H_n(X \times Y; \mathbb{F}) = \sum_{i+j=n} \beta_i(X; \mathbb{F}) \beta_j(Y; \mathbb{F}).$$

This is especially useful for coefficients in

$$\mathbb{Z}/2.$$

For example, if each factor has one copy of $\mathbb{Z}/2$ in degrees 0, 1, 2, 3, then the product has dimensions

$$1, 2, 3, 4, 3, 2, 1$$

in degrees

$$0, 1, 2, 3, 4, 5, 6.$$

3.9 The universal coefficient theorem for homology

The universal coefficient theorem for homology relates homology with integer coefficients to homology with coefficients in an abelian group G .

It gives a short exact sequence

$$0 \longrightarrow H_n(X) \otimes G \longrightarrow H_n(X; G) \longrightarrow \text{Tor}(H_{n-1}(X), G) \longrightarrow 0.$$

This sequence splits, although not naturally.

Thus, as an abstract group,

$$H_n(X; G) \cong (H_n(X) \otimes G) \oplus \text{Tor}(H_{n-1}(X), G).$$

For $G = \mathbb{Z}/2$, this becomes

$$H_n(X; \mathbb{Z}/2) \cong (H_n(X) \otimes \mathbb{Z}/2) \oplus \text{Tor}(H_{n-1}(X), \mathbb{Z}/2).$$

The first term

$$H_n(X) \otimes \mathbb{Z}/2$$

reduces ordinary n -dimensional homology classes modulo 2.

The second term

$$\text{Tor}(H_{n-1}(X), \mathbb{Z}/2)$$

detects 2-torsion in the previous integral homology group.

Thus mod-2 homology sees two things:

1. ordinary homology classes reduced modulo 2;
2. hidden 2-torsion from one dimension lower.

For example, if

$$H_1(X) \cong \mathbb{Z}/4 \oplus \mathbb{Z}/2,$$

then

$$H_1(X) \otimes \mathbb{Z}/2 \cong (\mathbb{Z}/2)^2.$$

Also,

$$\mathrm{Tor}(H_1(X), \mathbb{Z}/2) \cong (\mathbb{Z}/2)^2.$$

This is why mod-2 homology often has more generators than one might expect from the free part alone.

3.10 The universal coefficient theorem for cohomology

The universal coefficient theorem for cohomology says

$$0 \longrightarrow \mathrm{Ext}(H_{n-1}(X), \mathbb{Z}) \longrightarrow H^n(X; \mathbb{Z}) \longrightarrow \mathrm{Hom}(H_n(X), \mathbb{Z}) \longrightarrow 0.$$

This sequence also splits, although not naturally. Hence, as an abstract group,

$$H^n(X; \mathbb{Z}) \cong \mathrm{Hom}(H_n(X), \mathbb{Z}) \oplus \mathrm{Ext}(H_{n-1}(X), \mathbb{Z}).$$

We use the following facts:

$$\begin{aligned} \mathrm{Hom}(\mathbb{Z}, \mathbb{Z}) &\cong \mathbb{Z}, \\ \mathrm{Hom}(\mathbb{Z}^r, \mathbb{Z}) &\cong \mathbb{Z}^r, \\ \mathrm{Hom}(\mathbb{Z}/m, \mathbb{Z}) &= 0. \end{aligned}$$

The Hom term detects the free part of $H_n(X)$.

We also use

$$\begin{aligned} \mathrm{Ext}(\mathbb{Z}, \mathbb{Z}) &= 0, \\ \mathrm{Ext}(\mathbb{Z}^r, \mathbb{Z}) &= 0, \end{aligned}$$

and

$$\mathrm{Ext}(\mathbb{Z}/m, \mathbb{Z}) \cong \mathbb{Z}/m.$$

The Ext term detects torsion in $H_{n-1}(X)$.

Therefore, integral cohomology in degree n receives:

1. a free contribution from $H_n(X)$;
2. a torsion contribution from $H_{n-1}(X)$.

For example, if

$$H_2(X) \cong \mathbb{Z}/2$$

and

$$H_3(X) \cong \mathbb{Z}^2 \oplus \mathbb{Z}/2,$$

then

$$H^3(X) \cong \text{Hom}(H_3(X), \mathbb{Z}) \oplus \text{Ext}(H_2(X), \mathbb{Z}).$$

Now

$$\text{Hom}(\mathbb{Z}^2 \oplus \mathbb{Z}/2, \mathbb{Z}) \cong \mathbb{Z}^2,$$

and

$$\text{Ext}(\mathbb{Z}/2, \mathbb{Z}) \cong \mathbb{Z}/2.$$

So

$$H^3(X) \cong \mathbb{Z}^2 \oplus \mathbb{Z}/2.$$

3.11 Poincaré polynomial

Let X be a space whose homology groups over a field $\mathbb{F}$ are finite-dimensional. The k -th Betti number over $\mathbb{F}$ is

$$\beta_k^{\mathbb{F}}(X) = \dim_{\mathbb{F}} H_k(X; \mathbb{F}).$$

The Poincaré polynomial of X over $\mathbb{F}$ is

$$P_{\mathbb{F}}(X) = \sum_{k \geq 0} \beta_k^{\mathbb{F}}(X) t^k.$$

For example, if

$$H_0(X; \mathbb{F}) \cong \mathbb{F}, \quad H_1(X; \mathbb{F}) \cong \mathbb{F}^2, \quad H_2(X; \mathbb{F}) \cong \mathbb{F},$$

and all other homology groups vanish, then

$$P_{\mathbb{F}}(X) = 1 + 2t + t^2.$$

Evaluating at $t = -1$ gives

$$P_{\mathbb{F}}(X) \Big|_{t=-1} = \sum_k (-1)^k \beta_k^{\mathbb{F}}(X).$$

This alternating sum is equal to the Euler characteristic:

$$\chi(X) = P_{\mathbb{F}}(X) \Big|_{t=-1}.$$

3.12 Euler characteristic

Let X be a finite cell complex. If n_k is the number of k -cells of X , then the Euler characteristic of X is

$$\chi(X) = \sum_k (-1)^k n_k.$$

Equivalently,

$$\chi(X) = n_0 - n_1 + n_2 - n_3 + \cdots.$$

For a finite graph,

$$\chi(X) = V - E,$$

where V is the number of vertices and E is the number of edges.

For a closed triangulated surface,

$$\chi(X) = V - E + F.$$

For a finite CW complex, the Euler characteristic can also be computed from Betti numbers:

$$\chi(X) = \sum_k (-1)^k \beta_k^{\mathbb{F}}(X).$$

This number is independent of the field $\mathbb{F}$, even though the individual Betti numbers may depend on $\mathbb{F}$.

The reason is that the alternating sum of the dimensions of the chain groups equals the alternating sum of the dimensions of the homology groups:

$$\sum_k (-1)^k \dim_{\mathbb{F}} C_k(X; \mathbb{F}) = \sum_k (-1)^k \dim_{\mathbb{F}} H_k(X; \mathbb{F}).$$

Since

$$\dim_{\mathbb{F}} C_k(X; \mathbb{F}) = n_k,$$

we obtain

$$\chi(X) = \sum_k (-1)^k n_k = \sum_k (-1)^k \beta_k^{\mathbb{F}}(X).$$

3.13 Euler characteristic and products

If X and Y are finite cell complexes, then

$$\chi(X \times Y) = \chi(X)\chi(Y).$$

This can be proved directly from cells.

Suppose X has n_i cells in dimension i , and Y has m_j cells in dimension j . Then $X \times Y$ has

$$n_i m_j$$

cells of dimension $i + j$ coming from products of i -cells of X with j -cells of Y .

Therefore

$$\chi(X \times Y) = \sum_{i,j} (-1)^{i+j} n_i m_j.$$

Since

$$(-1)^{i+j} = (-1)^i (-1)^j,$$

we have

$$\chi(X \times Y) = \left(\sum_i (-1)^i n_i \right) \left(\sum_j (-1)^j m_j \right).$$

Hence

$$\chi(X \times Y) = \chi(X) \chi(Y).$$

Equivalently, using Poincaré polynomials over a field,

$$P_{\mathbb{F}}(X \times Y) = P_{\mathbb{F}}(X) P_{\mathbb{F}}(Y).$$

Evaluating at $t = -1$, we get

$$\chi(X \times Y) = \chi(X) \chi(Y).$$

3.14 Euler characteristic and inclusion-exclusion

If A and B are finite subcomplexes of a cell complex X , then

$$\chi(A \cup B) = \chi(A) + \chi(B) - \chi(A \cap B).$$

This is the Euler characteristic version of the inclusion-exclusion principle.

For each dimension k , let $n_k(A)$ be the number of k -cells of A , and similarly for B , $A \cup B$, and $A \cap B$. Then

$$n_k(A \cup B) = n_k(A) + n_k(B) - n_k(A \cap B).$$

Taking alternating sums gives

$$\sum_k (-1)^k n_k(A \cup B) = \sum_k (-1)^k n_k(A) + \sum_k (-1)^k n_k(B) - \sum_k (-1)^k n_k(A \cap B).$$

Therefore

$$\chi(A \cup B) = \chi(A) + \chi(B) - \chi(A \cap B).$$

This formula is especially useful when a space is built by gluing two subspaces together along a common intersection.

3.15 Quick reference table for the main algebraic tools

Tool	Formula
Tensor with $\mathbb{Z}$	$\mathbb{Z} \otimes G \cong G$
Tensor cyclic groups	$\mathbb{Z}/m \otimes \mathbb{Z}/n \cong \mathbb{Z}/\gcd(m, n)$
Tor of cyclic groups	$\text{Tor}(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n)$
Hom of free group	$\text{Hom}(\mathbb{Z}^r, \mathbb{Z}) \cong \mathbb{Z}^r$
Hom of torsion group	$\text{Hom}(\mathbb{Z}/m, \mathbb{Z}) = 0$
Ext of free group	$\text{Ext}(\mathbb{Z}^r, \mathbb{Z}) = 0$
Ext of cyclic group	$\text{Ext}(\mathbb{Z}/m, \mathbb{Z}) \cong \mathbb{Z}/m$

3.16 Quick reference table for the spaces in Problem 1

Space	H_0	H_1	H_2	H_3
L_4^3	$\mathbb{Z}$	$\mathbb{Z}/4$	0	$\mathbb{Z}$
$\mathbb{RP}^3$	$\mathbb{Z}$	$\mathbb{Z}/2$	0	$\mathbb{Z}$

With $\mathbb{Z}/2$ -coefficients, both spaces have

$$H_k(-; \mathbb{Z}/2) \cong \mathbb{Z}/2$$

for

$$k = 0, 1, 2, 3.$$

Thus

$$P_{\mathbb{Z}/2}(L_4^3) = 1 + t + t^2 + t^3,$$

and

$$P_{\mathbb{Z}/2}(\mathbb{RP}^3) = 1 + t + t^2 + t^3.$$

Therefore

$$P_{\mathbb{Z}/2}(L_4^3 \times \mathbb{RP}^3) = (1 + t + t^2 + t^3)^2.$$

Expanding,

$$(1 + t + t^2 + t^3)^2 = 1 + 2t + 3t^2 + 4t^3 + 3t^4 + 2t^5 + t^6.$$

Hence

$$\dim_{\mathbb{Z}/2} H_k(L_4^3 \times \mathbb{RP}^3; \mathbb{Z}/2) = 1, 2, 3, 4, 3, 2, 1$$

in degrees

$$0, 1, 2, 3, 4, 5, 6.$$

3.17 Conceptual summary

The two problems combine four central ideas.

First, cellular homology lets us compute $H_*(X)$ from the cells of X . For spaces such as

$$L_4^3 \quad \text{and} \quad \mathbb{RP}^3,$$

the cellular chain complexes are very small: each has one cell in dimensions 0, 1, 2, 3. The difference is in the attaching maps:

$$\partial_2 = 4$$

for L_4^3 , and

$$\partial_2 = 2$$

for $\mathbb{RP}^3$.

Second, the product CW structure allows us to build the cellular chain complex of

$$L_4^3 \times \mathbb{RP}^3.$$

The chain groups are obtained by combining cells whose dimensions add to the desired degree.

Third, the Künneth formula gives a more conceptual way to compute the homology of the product. It says that product homology contains tensor-product contributions and, over $\mathbb{Z}$, additional Tor contributions.

Fourth, the universal coefficient theorem explains how to pass from integral homology to cohomology and to homology with coefficients in $\mathbb{Z}/2$. In particular:

$$H^n(X)$$

depends on

$$H_n(X)$$

through Hom and on

$$H_{n-1}(X)$$

through Ext, while

$$H_n(X; \mathbb{Z}/2)$$

depends on

$$H_n(X)$$

through tensoring with $\mathbb{Z}/2$ and on

$$H_{n-1}(X)$$

through Tor.

Finally, the Euler characteristic is an alternating count of cells:

$$\chi(X) = \sum_k (-1)^k n_k.$$

It can also be computed as an alternating sum of Betti numbers:

$$\chi(X) = \sum_k (-1)^k \beta_k^{\mathbb{F}}(X) = P_{\mathbb{F}}(X) \Big|_{t=-1}.$$

It is multiplicative under products:

$$\chi(X \times Y) = \chi(X)\chi(Y),$$

and satisfies inclusion-exclusion:

$$\chi(A \cup B) = \chi(A) + \chi(B) - \chi(A \cap B).$$

4 Problem 3. Naturality of the Boundary Map for Cross Products

Statement

Let

$$\Phi_{(X,A)} : H^*(X, A) \otimes H^*(Y) \longrightarrow H^*(X \times Y, A \times Y)$$

be defined by

$$\Phi_{(X,A)}(\alpha \otimes \beta) = \alpha \times \beta.$$

Similarly, define

$$\Phi_A : H^*(A) \otimes H^*(Y) \longrightarrow H^*(A \times Y)$$

by

$$\Phi_A(\alpha \otimes \beta) = \alpha \times \beta.$$

Show that the cross product commutes with the connecting homomorphism in the long exact sequence of a pair. In the usual convention, this means

$$\delta_{(X \times Y, A \times Y)} \circ \Phi_A = \Phi_{(X,A)} \circ (\delta_{(X,A)} \otimes 1).$$

Equivalently, with the boundary notation used in the problem, this is written as

$$\delta \Phi_{(X,A)} = \Phi_A \delta.$$

Solution

Let

$$\alpha \in H^p(A), \quad \beta \in H^q(Y).$$

We want to show that

$$\delta(\alpha \times \beta) = (\delta\alpha) \times \beta.$$

Choose cocycle representatives

$$a \in C^p(A), \quad b \in C^q(Y),$$

with

$$da = 0, \quad db = 0.$$

To compute the connecting homomorphism

$$\delta_{(X,A)} : H^p(A) \rightarrow H^{p+1}(X, A),$$

extend a to a cochain

$$\tilde{a} \in C^p(X).$$

Then the relative cochain

$$d\tilde{a}$$

vanishes on A , because a was a cocycle on A . Hence

$$[d\tilde{a}] \in H^{p+1}(X, A),$$

and by definition

$$\delta[\alpha] = [d\tilde{a}].$$

Now consider the product pair

$$(X \times Y, A \times Y).$$

The cochain $a \times b$ on $A \times Y$ is extended by

$$\tilde{a} \times b$$

on $X \times Y$. Therefore

$$\delta(\alpha \times \beta) = [d(\tilde{a} \times b)].$$

Using the Leibniz rule for the coboundary of a cross product,

$$d(\tilde{a} \times b) = d\tilde{a} \times b + (-1)^p \tilde{a} \times db.$$

But b is a cocycle, so $db = 0$. Therefore

$$d(\tilde{a} \times b) = d\tilde{a} \times b.$$

Hence

$$\delta(\alpha \times \beta) = [d\tilde{a} \times b] = [d\tilde{a}] \times [b] = (\delta\alpha) \times \beta.$$

Thus the diagram

$$\begin{array}{ccc} H^p(A) \otimes H^q(Y) & \xrightarrow{\Phi_A} & H^{p+q}(A \times Y) \\ \delta \otimes 1 \downarrow & & \downarrow \delta \\ H^{p+1}(X, A) \otimes H^q(Y) & \xrightarrow{\Phi_{(X,A)}} & H^{p+q+1}(X \times Y, A \times Y) \end{array}$$

commutes.

Therefore

$$\boxed{\delta(\alpha \times \beta) = (\delta\alpha) \times \beta}$$

and so

$$\boxed{\delta\Phi_{(X,A)} = \Phi_A\delta}$$

in the notation of the problem.

5 Problem 4. Diagonal Maps and Cup Products

Statement

If X is a space, let

$$\Delta_X : X \rightarrow X \times X$$

be the diagonal map

$$\Delta_X(x) = (x, x).$$

Compute

$$\Delta_{S^2}^* : H^*(S^2 \times S^2) \rightarrow H^*(S^2)$$

and

$$\Delta_{T^2}^* : H^*(T^2 \times T^2) \rightarrow H^*(T^2).$$

Main idea

The diagonal map turns external products into cup products. Namely, if

$$\alpha \in H^p(X), \quad \beta \in H^q(X),$$

then

$$\Delta_X^*(\alpha \times \beta) = \alpha \smile \beta.$$

Indeed, the cup product is defined by

$$\alpha \smile \beta = \Delta_X^*(\alpha \times \beta).$$

Part I. The case of S^2

Let

$$u \in H^2(S^2; \mathbb{Z})$$

be a generator. Then

$$H^*(S^2; \mathbb{Z}) \cong \mathbb{Z}[u]/(u^2), \quad |u| = 2.$$

The relation $u^2 = 0$ holds because

$$H^4(S^2; \mathbb{Z}) = 0.$$

By the Künneth formula,

$$H^*(S^2 \times S^2; \mathbb{Z}) \cong H^*(S^2; \mathbb{Z}) \otimes H^*(S^2; \mathbb{Z}).$$

Let

$$u_1 = \pi_1^*(u), \quad u_2 = \pi_2^*(u),$$

where

$$\pi_1, \pi_2 : S^2 \times S^2 \rightarrow S^2$$

are the two projections.

Then

$$H^*(S^2 \times S^2; \mathbb{Z}) \cong \mathbb{Z}[u_1, u_2]/(u_1^2, u_2^2), \quad |u_1| = |u_2| = 2.$$

The diagonal map satisfies

$$\pi_1 \circ \Delta_{S^2} = \text{id}_{S^2}, \quad \pi_2 \circ \Delta_{S^2} = \text{id}_{S^2}.$$

Therefore

$$\Delta_{S^2}^*(u_1) = u, \quad \Delta_{S^2}^*(u_2) = u.$$

Since $\Delta_{S^2}^*$ is a ring homomorphism,

$$\Delta_{S^2}^*(u_1 u_2) = \Delta_{S^2}^*(u_1) \Delta_{S^2}^*(u_2) = u^2 = 0.$$

Hence

$$\boxed{\Delta_{S^2}^*(1) = 1}$$

$$\boxed{\Delta_{S^2}^*(u_1) = u, \quad \Delta_{S^2}^*(u_2) = u}$$

and

$$\boxed{\Delta_{S^2}^*(u_1 u_2) = 0.}$$

Equivalently, in degrees:

$$H^0(S^2 \times S^2) \cong \mathbb{Z} \longrightarrow H^0(S^2) \cong \mathbb{Z}$$

is the identity,

$$H^2(S^2 \times S^2) \cong \mathbb{Z}u_1 \oplus \mathbb{Z}u_2 \longrightarrow H^2(S^2) \cong \mathbb{Z}u$$

is given by

$$au_1 + bu_2 \longmapsto (a + b)u,$$

and

$$H^4(S^2 \times S^2) \cong \mathbb{Z}(u_1 u_2) \longrightarrow H^4(S^2) = 0$$

is the zero map.

Part II. The case of T^2

Let

$$T^2 = S^1 \times S^1.$$

The cohomology ring of the torus is

$$H^*(T^2; \mathbb{Z}) \cong \Lambda_{\mathbb{Z}}(\alpha, \beta),$$

where

$$|\alpha| = |\beta| = 1.$$

Thus

$$\begin{aligned} H^0(T^2) &\cong \mathbb{Z}, \\ H^1(T^2) &\cong \mathbb{Z}\alpha \oplus \mathbb{Z}\beta, \end{aligned}$$

and

$$H^2(T^2) \cong \mathbb{Z}(\alpha\beta).$$

Write

$$\omega = \alpha\beta.$$

Because the cup product is graded-commutative,

$$\beta\alpha = -\alpha\beta = -\omega.$$

Now consider

$$T^2 \times T^2.$$

Let

$$\alpha_1 = \pi_1^*(\alpha), \quad \beta_1 = \pi_1^*(\beta),$$

and

$$\alpha_2 = \pi_2^*(\alpha), \quad \beta_2 = \pi_2^*(\beta),$$

where

$$\pi_1, \pi_2 : T^2 \times T^2 \rightarrow T^2$$

are the two projections.

Since

$$\pi_1 \circ \Delta_{T^2} = \text{id}_{T^2}, \quad \pi_2 \circ \Delta_{T^2} = \text{id}_{T^2},$$

we get

$$\Delta_{T^2}^*(\alpha_1) = \alpha, \quad \Delta_{T^2}^*(\beta_1) = \beta,$$

and

$$\Delta_{T^2}^*(\alpha_2) = \alpha, \quad \Delta_{T^2}^*(\beta_2) = \beta.$$

Therefore $\Delta_{T^2}^*$ is determined by

$$\boxed{\alpha_1 \mapsto \alpha, \quad \beta_1 \mapsto \beta, \quad \alpha_2 \mapsto \alpha, \quad \beta_2 \mapsto \beta.}$$

Since $\Delta_{T^2}^*$ is a ring homomorphism, we can compute all products.

For degree 2, we obtain:

$$\Delta_{T^2}^*(\alpha_1\beta_1) = \alpha\beta = \omega,$$

$$\Delta_{T^2}^*(\alpha_2\beta_2) = \alpha\beta = \omega,$$

$$\Delta_{T^2}^*(\alpha_1\alpha_2) = \alpha\alpha = 0,$$

$$\Delta_{T^2}^*(\beta_1\beta_2) = \beta\beta = 0,$$

$$\Delta_{T^2}^*(\alpha_1\beta_2) = \alpha\beta = \omega,$$

and

$$\Delta_{T^2}^*(\beta_1\alpha_2) = \beta\alpha = -\omega.$$

Thus

$$\boxed{\Delta_{T^2}^*(\alpha_1\beta_1) = \omega}$$

$$\boxed{\Delta_{T^2}^*(\alpha_2\beta_2) = \omega}$$

$$\boxed{\Delta_{T^2}^*(\alpha_1\alpha_2) = 0}$$

$$\boxed{\Delta_{T^2}^*(\beta_1\beta_2) = 0}$$

$$\boxed{\Delta_{T^2}^*(\alpha_1\beta_2) = \omega}$$

and

$$\boxed{\Delta_{T^2}^*(\beta_1\alpha_2) = -\omega.}$$

All classes of degree 3 and 4 in $H^*(T^2 \times T^2)$ map to zero because

$$H^3(T^2) = 0$$

and

$$H^4(T^2) = 0.$$

Therefore the diagonal map sends the generators from both factors to the same generators on T^2 , and then multiplies them using the ordinary cup product on T^2 .

6 Problem 5. Relative Cup Products and Open Covers by Contractible Sets

Statement

Let $U, V \subset X$ be open sets. If

$$x \in H^*(X, U)$$

and

$$y \in H^*(X, V),$$

show that

$$x \smile y \in H^*(X, U \cup V).$$

Using this, show that if X has a covering by n contractible open subsets, then

$$a_1 \smile a_2 \smile \cdots \smile a_n = 0$$

whenever

$$a_1, \dots, a_n \in H^*(X)$$

have positive degree.

Deduce that if T^k can be covered by n contractible open subsets, then

$$n > k.$$

Finally, find a covering of T^2 by three contractible open subsets.

Part I. The relative cup product

Let

$$x \in H^p(X, U), \quad y \in H^q(X, V).$$

The external product gives

$$x \times y \in H^{p+q}(X \times X, U \times X \cup X \times V).$$

Now consider the diagonal map

$$\Delta : X \rightarrow X \times X, \quad \Delta(z) = (z, z).$$

If

$$z \in U \cup V,$$

then either $z \in U$ or $z \in V$. Hence

$$(z, z) \in U \times X$$

or

$$(z, z) \in X \times V.$$

Therefore

$$\Delta(U \cup V) \subset U \times X \cup X \times V.$$

Thus Δ is a map of pairs

$$\Delta : (X, U \cup V) \longrightarrow (X \times X, U \times X \cup X \times V).$$

Pulling back the external product by Δ , we obtain

$$\Delta^*(x \times y) \in H^{p+q}(X, U \cup V).$$

By definition, this is the relative cup product:

$$x \smile y = \Delta^*(x \times y).$$

Therefore

$$\boxed{x \smile y \in H^{p+q}(X, U \cup V).}$$

Part II. Vanishing of products from a contractible open cover

Assume that

$$X = U_1 \cup U_2 \cup \cdots \cup U_n,$$

where each U_i is contractible and open.

Let

$$a_i \in H^{d_i}(X)$$

with

$$d_i > 0.$$

Since U_i is contractible,

$$H^{d_i}(U_i) = 0$$

for $d_i > 0$. Therefore the restriction of a_i to U_i is zero:

$$a_i|_{U_i} = 0.$$

From the long exact sequence of the pair (X, U_i) ,

$$\cdots \rightarrow H^{d_i}(X, U_i) \rightarrow H^{d_i}(X) \rightarrow H^{d_i}(U_i) \rightarrow \cdots,$$

we conclude that a_i comes from a relative cohomology class. Thus there exists

$$\tilde{a}_i \in H^{d_i}(X, U_i)$$

such that its image in $H^{d_i}(X)$ is a_i .

Now use the relative cup product repeatedly:

$$\tilde{a}_1 \smile \tilde{a}_2 \in H^*(X, U_1 \cup U_2),$$

then

$$\tilde{a}_1 \smile \tilde{a}_2 \smile \tilde{a}_3 \in H^*(X, U_1 \cup U_2 \cup U_3),$$

and so on. Finally,

$$\tilde{a}_1 \smile \cdots \smile \tilde{a}_n \in H^*(X, U_1 \cup \cdots \cup U_n).$$

But

$$U_1 \cup \cdots \cup U_n = X.$$

Hence

$$H^*(X, U_1 \cup \cdots \cup U_n) = H^*(X, X) = 0.$$

Therefore

$$\tilde{a}_1 \smile \cdots \smile \tilde{a}_n = 0.$$

Passing to absolute cohomology gives

$$a_1 \smile a_2 \smile \cdots \smile a_n = 0.$$

Thus

$$\boxed{a_1 \smile a_2 \smile \cdots \smile a_n = 0}$$

for all positive-degree classes $a_1, \dots, a_n \in H^*(X)$.

Part III. Application to the torus T^k

The cohomology ring of the k -torus is

$$H^*(T^k; \mathbb{Z}) \cong \Lambda_{\mathbb{Z}}(\alpha_1, \dots, \alpha_k),$$

where

$$|\alpha_i| = 1.$$

The product

$$\alpha_1 \smile \alpha_2 \smile \cdots \smile \alpha_k$$

is the orientation class of T^k . In particular,

$$\alpha_1 \smile \cdots \smile \alpha_k \neq 0.$$

Moreover, for every $r \leq k$,

$$\alpha_1 \smile \cdots \smile \alpha_r \neq 0.$$

Suppose T^k could be covered by n contractible open subsets with

$$n \leq k.$$

Then by the result above, every product of n positive-degree cohomology classes would have to vanish.

But

$$\alpha_1 \smile \cdots \smile \alpha_n \neq 0.$$

This is a contradiction.

Therefore

$$\boxed{n > k.}$$

So T^k cannot be covered by k or fewer contractible open subsets.

Part IV. A covering of T^2 by three contractible open subsets

We represent T^2 as the quotient of a square by identifying opposite sides:

$$T^2 = [0, 1] \times [0, 1] / \sim,$$

where

$$(0, y) \sim (1, y)$$

and

$$(x, 0) \sim (x, 1).$$

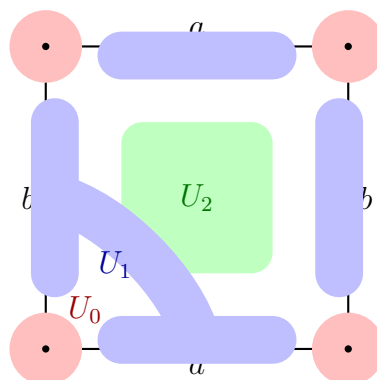
The square model has one 0-cell, two 1-cells, and one 2-cell. We construct three open subsets.

1. U_0 : a small open neighborhood of the unique vertex of the quotient. In the square picture, this appears as four small neighborhoods of the four corners. After identifying the corners, these four pieces become one open disk. Hence U_0 is contractible.
2. U_2 : a large open disk contained in the interior of the square. This covers the middle of the 2-cell. Since it is an open disk, U_2 is contractible.
3. U_1 : an open regular neighborhood of a tree joining the middle parts of the two 1-cells. In the square picture, this appears as thin strips near the middle portions of the boundary edges, together with a small bridge through the interior. Since a regular neighborhood of a tree in a surface is an open disk, U_1 is contractible.

Choosing the neighborhoods large enough, these three open sets cover the whole torus:

$$T^2 = U_0 \cup U_1 \cup U_2.$$

The following picture is schematic. Opposite edges of the square are identified.



After quotienting the square to form the torus:

$U_0 \simeq \text{open disk},$	$U_1 \simeq \text{open disk},$	$U_2 \simeq \text{open disk}.$
--------------------------------	--------------------------------	--------------------------------

Hence each U_i is contractible, and

$$T^2 = U_0 \cup U_1 \cup U_2.$$

This agrees with the lower bound from cup products. Since

$$H^*(T^2) \cong \Lambda(\alpha, \beta)$$

and

$$\alpha \smile \beta \neq 0,$$

the torus cannot be covered by two contractible open sets. But the construction above shows that it can be covered by three. Therefore the minimal number is exactly three:

$$T^2 \text{ can be covered by 3 contractible open subsets, but not by 2.}$$

7 Theory Behind Cross Products, Diagonal Maps, Relative Cup Products, and Contractible Covers

7.1 The common idea behind Problems 3–5

The three problems are all consequences of one fundamental principle:

$$\text{cup products are obtained from cross products by pulling back along the diagonal map.}$$

More precisely, if X is a topological space and

$$\Delta_X : X \longrightarrow X \times X, \quad \Delta_X(x) = (x, x),$$

is the diagonal map, then for cohomology classes

$$\alpha \in H^p(X), \quad \beta \in H^q(X),$$

the cup product is defined by

$$\alpha \smile \beta = \Delta_X^*(\alpha \times \beta).$$

Thus the basic mechanism is

$$\text{cross product} \quad + \quad \text{diagonal pullback} \quad = \quad \text{cup product.}$$

In diagram form:

$$H^p(X) \otimes H^q(X) \xrightarrow{\times} H^{p+q}(X \times X) \xrightarrow{\Delta_X^*} H^{p+q}(X)$$

and the resulting composite is precisely

$$\alpha \otimes \beta \longmapsto \alpha \smile \beta.$$

The three problems use this idea in different ways:

- Problem 3 studies how cross products behave with the connecting homomorphism in relative cohomology.
- Problem 4 computes diagonal pullbacks explicitly for S^2 and T^2 .
- Problem 5 uses relative cup products to prove lower bounds for coverings by contractible open sets.

7.2 Cohomology is contravariant

A continuous map

$$f : X \rightarrow Y$$

induces a map in homology

$$f_* : H_n(X) \rightarrow H_n(Y).$$

Thus homology is covariant.

Cohomology goes in the opposite direction. The same continuous map

$$f : X \rightarrow Y$$

induces

$$f^* : H^n(Y) \rightarrow H^n(X).$$

Thus cohomology is contravariant.

This reversal of arrows is essential for the diagonal map. Since

$$\Delta_X : X \rightarrow X \times X,$$

contravariance gives

$$\Delta_X^* : H^*(X \times X) \rightarrow H^*(X).$$

So the diagonal map allows us to take a class on $X \times X$ and pull it back to a class on X .

This is exactly what is needed in the definition of the cup product:

$$\alpha \times \beta \in H^{p+q}(X \times X),$$

but

$$\alpha \smile \beta \in H^{p+q}(X).$$

The diagonal pullback performs the passage

$$H^{p+q}(X \times X) \longrightarrow H^{p+q}(X).$$

7.3 The cohomological cross product

Let X and Y be topological spaces. The cohomological cross product is a bilinear map

$$\times : H^p(X) \otimes H^q(Y) \longrightarrow H^{p+q}(X \times Y).$$

Given

$$\alpha \in H^p(X), \quad \beta \in H^q(Y),$$

their cross product is written

$$\alpha \times \beta \in H^{p+q}(X \times Y).$$

Intuitively, the class

$$\alpha \times \beta$$

measures α in the X -direction and β in the Y -direction.

For example, if

$$X = S^2, \quad Y = S^2,$$

and

$$u \in H^2(S^2; \mathbb{Z})$$

is the fundamental cohomology class of S^2 , then on

$$S^2 \times S^2$$

we get two natural degree 2 classes:

$$u_1 = u \times 1, \quad u_2 = 1 \times u.$$

Equivalently,

$$u_1 = \pi_1^*(u), \quad u_2 = \pi_2^*(u),$$

where

$$\pi_1 : S^2 \times S^2 \rightarrow S^2, \quad \pi_2 : S^2 \times S^2 \rightarrow S^2$$

are the two projections.

The class u_1 comes from the first sphere, while u_2 comes from the second sphere.

7.4 The cochain-level Leibniz rule

At the cochain level, the cross product satisfies a graded Leibniz rule.

If

$$a \in C^p(X), \quad b \in C^q(Y),$$

then

$$a \times b \in C^{p+q}(X \times Y),$$

and the coboundary satisfies

$$\boxed{d(a \times b) = da \times b + (-1)^p a \times db.}$$

This is the cohomological analogue of the product rule

$$d(fg) = df \cdot g + f \cdot dg.$$

The sign

$$(-1)^p$$

appears because cochains are graded objects.

If b is a cocycle, then

$$db = 0.$$

Therefore the formula becomes

$$d(a \times b) = da \times b.$$

This simple identity is the reason why the cross product is compatible with the connecting homomorphism in the long exact sequence of a pair.

7.5 Relative cohomology and the connecting homomorphism

Let $A \subset X$. The relative cohomology group

$$H^n(X, A)$$

measures cohomology of X relative to A .

A useful intuition is

$$H^n(X, A) \cong \widetilde{H}^n(X/A)$$

under suitable hypotheses. Thus, morally, relative cohomology treats A as collapsed or ignored.

The long exact sequence of a pair is

$$\cdots \longrightarrow H^n(X, A) \longrightarrow H^n(X) \longrightarrow H^n(A) \xrightarrow{\delta} H^{n+1}(X, A) \longrightarrow H^{n+1}(X) \longrightarrow \cdots .$$

The connecting homomorphism

$$\delta : H^n(A) \rightarrow H^{n+1}(X, A)$$

measures the obstruction to extending a cohomology class on A to a cohomology class on X .

Indeed, if

$$\alpha \in H^n(A)$$

extends to a class

$$\tilde{\alpha} \in H^n(X),$$

then

$$\delta\alpha = 0.$$

If α does not extend, then

$$\delta\alpha$$

records the obstruction.

7.6 Cross products and relative boundary maps

Let (X, A) be a pair and let Y be another space. The cross product gives maps

$$H^p(A) \otimes H^q(Y) \longrightarrow H^{p+q}(A \times Y)$$

and

$$H^{p+1}(X, A) \otimes H^q(Y) \longrightarrow H^{p+q+1}(X \times Y, A \times Y).$$

The naturality statement is that the following diagram commutes:

$$\begin{array}{ccc} H^p(A) \otimes H^q(Y) & \xrightarrow{\quad \times \quad} & H^{p+q}(A \times Y) \\ \delta \otimes 1 \downarrow & & \downarrow \delta \\ H^{p+1}(X, A) \otimes H^q(Y) & \xrightarrow{\quad \times \quad} & H^{p+q+1}(X \times Y, A \times Y) \end{array}$$

Equivalently,

$$\boxed{\delta(\alpha \times \beta) = (\delta\alpha) \times \beta.}$$

Here

$$\alpha \in H^p(A), \quad \beta \in H^q(Y).$$

Cochain explanation

Choose cocycle representatives

$$a \in C^p(A), \quad b \in C^q(Y),$$

with

$$da = 0, \quad db = 0.$$

To compute

$$\delta[\alpha] \in H^{p+1}(X, A),$$

extend a to a cochain

$$\tilde{a} \in C^p(X).$$

Then

$$d\tilde{a}$$

vanishes on A , because a was already a cocycle on A . Thus

$$[d\tilde{a}] \in H^{p+1}(X, A)$$

represents

$$\delta[\alpha].$$

Now consider the product

$$a \times b$$

on

$$A \times Y.$$

It is extended to $X \times Y$ by

$$\tilde{a} \times b.$$

Therefore

$$\delta(\alpha \times \beta) = [d(\tilde{a} \times b)].$$

Using the Leibniz rule,

$$d(\tilde{a} \times b) = d\tilde{a} \times b + (-1)^p \tilde{a} \times db.$$

Since b is a cocycle,

$$db = 0.$$

Hence

$$d(\tilde{a} \times b) = d\tilde{a} \times b.$$

Therefore

$$\delta(\alpha \times \beta) = [d\tilde{a} \times b] = [d\tilde{a}] \times [b] = (\delta\alpha) \times \beta.$$

Thus

$$\boxed{\delta(\alpha \times \beta) = (\delta\alpha) \times \beta.}$$

7.7 The diagonal map and the cup product

For any topological space X , the diagonal map is

$$\Delta_X : X \rightarrow X \times X, \quad \Delta_X(x) = (x, x).$$

Geometrically, Δ_X embeds X into $X \times X$ as the set of equal pairs:

$$\Delta_X(X) = \{(x, x) : x \in X\}.$$

For example, if $X = \mathbb{R}$, then

$$\Delta_{\mathbb{R}}(t) = (t, t)$$

is the ordinary diagonal line inside $\mathbb{R}^2$.

In cohomology, the diagonal map gives

$$\Delta_X^* : H^*(X \times X) \rightarrow H^*(X).$$

The cup product is defined by

$$\boxed{\alpha \smile \beta = \Delta_X^*(\alpha \times \beta)}.$$

Thus the cup product is not an independent mysterious operation. It is the external cross product restricted back to X using the diagonal.

$$\alpha, \beta \in H^*(X) \xrightarrow{\text{cross product}} \alpha \times \beta \in H^*(X \times X) \xrightarrow{\Delta_X^*} \alpha \smile \beta \in H^*(X)$$

7.8 Diagonal maps on S^2

Let

$$u \in H^2(S^2; \mathbb{Z})$$

be a generator. The cohomology ring of S^2 is

$$H^*(S^2; \mathbb{Z}) \cong \mathbb{Z}[u]/(u^2), \quad |u| = 2.$$

The relation

$$u^2 = 0$$

does not mean that $u = 0$. It means that

$$u^2 \in H^4(S^2; \mathbb{Z}),$$

but

$$H^4(S^2; \mathbb{Z}) = 0.$$

So the square of the degree 2 generator must vanish.

Now consider

$$S^2 \times S^2.$$

Let

$$u_1 = \pi_1^*(u), \quad u_2 = \pi_2^*(u),$$

where

$$\pi_1, \pi_2 : S^2 \times S^2 \rightarrow S^2$$

are the projections.

By the Künneth formula,

$$H^*(S^2 \times S^2; \mathbb{Z}) \cong \mathbb{Z}[u_1, u_2]/(u_1^2, u_2^2), \quad |u_1| = |u_2| = 2.$$

The diagonal map

$$\Delta_{S^2} : S^2 \rightarrow S^2 \times S^2$$

satisfies

$$\pi_1 \circ \Delta_{S^2} = \text{id}_{S^2}, \quad \pi_2 \circ \Delta_{S^2} = \text{id}_{S^2}.$$

Therefore

$$\Delta_{S^2}^*(u_1) = u, \quad \Delta_{S^2}^*(u_2) = u.$$

Since $\Delta_{S^2}^*$ is a ring homomorphism,

$$\Delta_{S^2}^*(u_1 u_2) = \Delta_{S^2}^*(u_1) \Delta_{S^2}^*(u_2) = u^2 = 0.$$

Thus the complete behavior is

$$\boxed{\Delta_{S^2}^*(1) = 1,}$$

$$\Delta_{S^2}^*(u_1) = u, \quad \Delta_{S^2}^*(u_2) = u,$$

and

$$\Delta_{S^2}^*(u_1 u_2) = 0.$$

In degree 2, this means

$$au_1 + bu_2 \mapsto (a + b)u.$$

In degree 4, the generator

$$u_1 u_2 \in H^4(S^2 \times S^2)$$

maps to zero because

$$H^4(S^2) = 0.$$

7.9 Diagonal maps on T^2

The torus is

$$T^2 = S^1 \times S^1.$$

Its cohomology ring is

$$H^*(T^2; \mathbb{Z}) \cong \Lambda_{\mathbb{Z}}(\alpha, \beta),$$

where

$$|\alpha| = |\beta| = 1.$$

This means that

$$\alpha^2 = 0, \quad \beta^2 = 0, \quad \alpha\beta = -\beta\alpha.$$

Let

$$\omega = \alpha\beta.$$

Then

$$\omega \in H^2(T^2; \mathbb{Z})$$

is the orientation class of the torus.

Geometrically:

- α detects winding in one circle direction;
- β detects winding in the other circle direction;
- $\alpha\beta$ detects the two-dimensional fundamental class.

Now consider

$$T^2 \times T^2.$$

Let

$$\alpha_1 = \pi_1^*(\alpha), \quad \beta_1 = \pi_1^*(\beta),$$

and

$$\alpha_2 = \pi_2^*(\alpha), \quad \beta_2 = \pi_2^*(\beta).$$

The diagonal map

$$\Delta_{T^2} : T^2 \rightarrow T^2 \times T^2$$

satisfies

$$\pi_1 \circ \Delta_{T^2} = \text{id}_{T^2}, \quad \pi_2 \circ \Delta_{T^2} = \text{id}_{T^2}.$$

Therefore

$$\Delta_{T^2}^*(\alpha_1) = \alpha, \quad \Delta_{T^2}^*(\beta_1) = \beta,$$

and

$$\Delta_{T^2}^*(\alpha_2) = \alpha, \quad \Delta_{T^2}^*(\beta_2) = \beta.$$

Hence

$$\boxed{\alpha_1 \mapsto \alpha, \quad \beta_1 \mapsto \beta, \quad \alpha_2 \mapsto \alpha, \quad \beta_2 \mapsto \beta.}$$

All products are computed by replacing each subscripted generator by its unsubscripted version and using the exterior algebra rules.

For example,

$$\Delta_{T^2}^*(\alpha_1\beta_2) = \alpha\beta = \omega.$$

Also,

$$\Delta_{T^2}^*(\alpha_1\alpha_2) = \alpha\alpha = 0.$$

The sign in

$$\Delta_{T^2}^*(\beta_1\alpha_2)$$

comes from graded commutativity:

$$\beta\alpha = (-1)^{1 \cdot 1}\alpha\beta = -\alpha\beta.$$

Thus

$$\Delta_{T^2}^*(\beta_1\alpha_2) = -\omega.$$

The important degree 2 computations are

$$\Delta_{T^2}^*(\alpha_1\beta_1) = \omega,$$

$$\Delta_{T^2}^*(\alpha_2\beta_2) = \omega,$$

$$\Delta_{T^2}^*(\alpha_1\alpha_2) = 0,$$

$$\Delta_{T^2}^*(\beta_1\beta_2) = 0,$$

$$\Delta_{T^2}^*(\alpha_1\beta_2) = \omega,$$

and

$$\Delta_{T^2}^*(\beta_1\alpha_2) = -\omega.$$

All degree 3 and degree 4 classes in $H^*(T^2 \times T^2)$ map to zero because

$$H^3(T^2) = 0$$

and

$$H^4(T^2) = 0.$$

7.10 Relative cup products

Let $U, V \subset X$ be open subsets.

Suppose

$$x \in H^p(X, U), \quad y \in H^q(X, V).$$

The relative cup product satisfies

$$x \smile y \in H^{p+q}(X, U \cup V).$$

The reason is intuitive.

A class in $H^p(X, U)$ vanishes on U , while a class in $H^q(X, V)$ vanishes on V . Therefore their product vanishes wherever either factor vanishes, namely on

$$U \cup V.$$

At the level of cross products, we first form

$$x \times y \in H^{p+q}(X \times X, U \times X \cup X \times V).$$

Now consider the diagonal map

$$\Delta : X \rightarrow X \times X, \quad \Delta(z) = (z, z).$$

If

$$z \in U \cup V,$$

then either $z \in U$ or $z \in V$.

If $z \in U$, then

$$(z, z) \in U \times X.$$

If $z \in V$, then

$$(z, z) \in X \times V.$$

Hence

$$\Delta(U \cup V) \subset U \times X \cup X \times V.$$

Therefore the diagonal is a map of pairs

$$\Delta : (X, U \cup V) \longrightarrow (X \times X, U \times X \cup X \times V).$$

Pulling back the external product gives

$$\Delta^*(x \times y) \in H^{p+q}(X, U \cup V).$$

By definition,

$$x \smile y = \Delta^*(x \times y).$$

Thus

$$x \smile y \in H^{p+q}(X, U \cup V).$$

7.11 Contractible open covers and vanishing cup products

Suppose that X has a cover by n contractible open sets:

$$X = U_1 \cup U_2 \cup \cdots \cup U_n.$$

Let

$$a_1, \dots, a_n \in H^*(X)$$

be positive-degree cohomology classes.

Write

$$a_i \in H^{d_i}(X), \quad d_i > 0.$$

Because each U_i is contractible,

$$H^{d_i}(U_i) = 0$$

for $d_i > 0$. Therefore the restriction of a_i to U_i is zero:

$$a_i|_{U_i} = 0.$$

Now use the long exact sequence of the pair (X, U_i) :

$$\cdots \longrightarrow H^{d_i}(X, U_i) \longrightarrow H^{d_i}(X) \longrightarrow H^{d_i}(U_i) \longrightarrow \cdots.$$

Since

$$a_i|_{U_i} = 0,$$

the class a_i comes from a relative class

$$\tilde{a}_i \in H^{d_i}(X, U_i).$$

Now multiply these relative classes:

$$\tilde{a}_1 \smile \tilde{a}_2 \in H^*(X, U_1 \cup U_2),$$

then

$$\tilde{a}_1 \smile \tilde{a}_2 \smile \tilde{a}_3 \in H^*(X, U_1 \cup U_2 \cup U_3),$$

and continuing,

$$\tilde{a}_1 \smile \cdots \smile \tilde{a}_n \in H^*(X, U_1 \cup \cdots \cup U_n).$$

But

$$U_1 \cup \cdots \cup U_n = X.$$

Therefore

$$H^*(X, U_1 \cup \cdots \cup U_n) = H^*(X, X) = 0.$$

Hence

$$\tilde{a}_1 \smile \cdots \smile \tilde{a}_n = 0.$$

Passing to absolute cohomology gives

$$\boxed{a_1 \smile a_2 \smile \cdots \smile a_n = 0.}$$

Thus we have proved the important principle:

If X is covered by n contractible open sets, then every n -fold product of positive-degree classes vanishes.

7.12 Cup-length

The cup-length of a space X is the largest integer r such that there exist positive-degree cohomology classes

$$a_1, \dots, a_r \in H^*(X)$$

with

$$a_1 \smile a_2 \smile \dots \smile a_r \neq 0.$$

Thus

$$\text{cup-length}(X) = \max \{r : \exists a_1, \dots, a_r, |a_i| > 0, a_1 \cdots a_r \neq 0\}.$$

The previous theorem implies:

If X can be covered by n contractible open sets, then $n > \text{cup-length}(X)$.
--

Equivalently,

$n \geq \text{cup-length}(X) + 1.$

This is closely related to the Lusternik–Schnirelmann category of a space. In many contexts, the cup-length gives a lower bound for how many contractible pieces are needed to cover a space.

7.13 The cup-length of the torus T^k

The k -dimensional torus is

$$T^k = (S^1)^k.$$

Its cohomology ring is

$$H^*(T^k; \mathbb{Z}) \cong \Lambda_{\mathbb{Z}}(\alpha_1, \dots, \alpha_k),$$

where

$$|\alpha_i| = 1.$$

This is an exterior algebra, so

$$\alpha_i^2 = 0$$

for every i , and

$$\alpha_i \alpha_j = -\alpha_j \alpha_i$$

for $i \neq j$.

The product

$$\alpha_1 \smile \alpha_2 \smile \cdots \smile \alpha_k$$

is nonzero. It is the top-dimensional cohomology class, also called the orientation class of T^k .

Therefore

$$\text{cup-length}(T^k) = k.$$

Hence, if T^k can be covered by n contractible open subsets, then

$$n > \text{cup-length}(T^k) = k.$$

So

$$\boxed{n > k.}$$

Equivalently,

$$\boxed{n \geq k + 1.}$$

For the ordinary torus T^2 , this gives

$$n \geq 3.$$

Thus T^2 cannot be covered by two contractible open subsets.

7.14 Why the torus T^2 needs three contractible open sets

The torus has two independent degree 1 cohomology classes

$$\alpha, \beta \in H^1(T^2; \mathbb{Z}),$$

and their product is nonzero:

$$\alpha \smile \beta \neq 0.$$

This product lies in

$$H^2(T^2; \mathbb{Z}) \cong \mathbb{Z}$$

and represents the orientation class of the torus.

If T^2 could be covered by two contractible open sets,

$$T^2 = U \cup V,$$

then the vanishing theorem would imply that every product of two positive-degree cohomology classes must vanish.

In particular, it would imply

$$\alpha \smile \beta = 0.$$

But this is false.

Therefore T^2 cannot be covered by two contractible open sets.

Since one can construct a cover by three contractible open sets, the minimal number is exactly

$$\boxed{3}.$$

7.15 A geometric picture of a three-set cover of T^2

Represent the torus as the quotient of a square by identifying opposite sides:

$$T^2 = [0, 1] \times [0, 1] / \sim,$$

where

$$(0, y) \sim (1, y), \quad (x, 0) \sim (x, 1).$$

The square model has:

$$1 \text{ vertex}, \quad 2 \text{ edge loops}, \quad 1 \text{ face}.$$

A natural cover by three contractible open sets is:

$$U_0 = \text{a small neighborhood of the identified vertex},$$

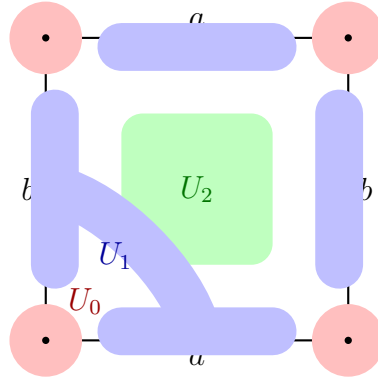
$$U_1 = \text{a regular neighborhood of a tree-like part of the 1-skeleton},$$

$$U_2 = \text{a large open disk inside the 2-cell}.$$

The key point is that U_1 must be chosen around a tree, not around the full 1-skeleton. A neighborhood of the full 1-skeleton of the torus would contain loops and would not be

contractible. A tree has no cycles, and a regular neighborhood of a tree in a surface is an open disk.

The following diagram is schematic. Opposite sides of the square are identified.



After the quotient identifications:

$$U_0 \simeq \text{open disk}, \quad U_1 \simeq \text{open disk}, \quad U_2 \simeq \text{open disk}.$$

Thus each U_i is contractible, and

$$T^2 = U_0 \cup U_1 \cup U_2.$$

Therefore

$T^2 \text{ can be covered by three contractible open sets, but not by two.}$

7.16 Summary of the theory

The logical chain behind Problems 3–5 is:

relative cohomology $\implies$ relative cross product $\implies$ diagonal pullback $\implies$ relative cup product $\implies$ vanishing

Problem 3 proves that cross products behave well with connecting homomorphisms:

$$\delta(\alpha \times \beta) = (\delta\alpha) \times \beta.$$

Problem 4 uses the diagonal map to compute cup products:

$$\alpha \smile \beta = \Delta_X^*(\alpha \times \beta).$$

Problem 5 uses relative cup products to show that if

$$X = U_1 \cup \cdots \cup U_n$$

is a cover by contractible open sets, then every n -fold cup product of positive-degree cohomology classes must vanish:

$$a_1 \smile \cdots \smile a_n = 0.$$

For the torus,

$$H^*(T^k; \mathbb{Z}) \cong \Lambda_{\mathbb{Z}}(\alpha_1, \dots, \alpha_k),$$

and

$$\alpha_1 \smile \cdots \smile \alpha_k \neq 0.$$

Therefore the torus has cup-length k , and any cover of T^k by contractible open sets must have more than k members:

$$\boxed{n > k.}$$

For T^2 , the exact minimum is

$$\boxed{3.}$$

8 Concrete Examples for Cross Products, Diagonal Maps, Cup Products, and Contractible Covers

8.1 Overview of the examples

The purpose of this section is to make the theory from Problems 3–5 more concrete. The main ideas are:

$$\text{cross product} + \text{diagonal pullback} = \text{cup product},$$

and

$$\text{nonzero cup products} \implies \text{lower bounds on contractible open covers.}$$

Unless stated otherwise, all cohomology groups are taken with coefficients in $\mathbb{Z}$.

We organize the examples into three levels:

- easy examples, where the computations show the basic mechanism;
- medium examples, where the cup product gives useful covering obstructions;
- deeper examples, where the cohomology ring structure becomes essential.

8.2 Easy Examples

Example 1. The diagonal map on S^1

Let

$$X = S^1.$$

The cohomology ring of the circle is

$$H^*(S^1) \cong \Lambda(a), \quad |a| = 1.$$

Thus

$$H^0(S^1) \cong \mathbb{Z}, \quad H^1(S^1) \cong \mathbb{Z}a,$$

and

$$a^2 = 0.$$

Now consider

$$S^1 \times S^1 = T^2.$$

Let

$$a_1 = \pi_1^*(a), \quad a_2 = \pi_2^*(a),$$

where

$$\pi_1 : S^1 \times S^1 \rightarrow S^1, \quad \pi_2 : S^1 \times S^1 \rightarrow S^1$$

are the two projections. Then

$$H^*(S^1 \times S^1) \cong \Lambda(a_1, a_2).$$

The diagonal map is

$$\Delta : S^1 \rightarrow S^1 \times S^1, \quad \Delta(x) = (x, x).$$

Since

$$\pi_1 \circ \Delta = \text{id}_{S^1}, \quad \pi_2 \circ \Delta = \text{id}_{S^1},$$

we get

$$\Delta^*(a_1) = a, \quad \Delta^*(a_2) = a.$$

Now compute the top product on $S^1 \times S^1$:

$$\Delta^*(a_1 a_2) = \Delta^*(a_1) \Delta^*(a_2) = aa = a^2 = 0.$$

Therefore

$$\boxed{\Delta^*(a_1) = a,}$$

$$\boxed{\Delta^*(a_2) = a,}$$

and

$$\boxed{\Delta^*(a_1 a_2) = 0.}$$

This is the simplest model of the diagonal map. The diagonal identifies the two circle directions, so both a_1 and a_2 become the same class a . But their product becomes a^2 , which is zero because S^1 has no degree 2 cohomology.

Example 2. Why S^1 cannot be covered by one contractible open set

The circle has

$$H^1(S^1) \cong \mathbb{Z}.$$

Let

$$a \in H^1(S^1)$$

be a generator. Then

$$a \neq 0.$$

The cup-length of S^1 is 1, because there exists one nonzero positive-degree class, but every product of two positive-degree classes is zero:

$$a^2 = 0.$$

Thus

$$\text{cup-length}(S^1) = 1.$$

The vanishing theorem for contractible open covers says:

if X is covered by n contractible open sets, then every n -fold product of positive-degree classes vanishes.

If S^1 could be covered by one contractible open set, then every one-fold positive-degree class would vanish. But

$$a \neq 0.$$

Therefore S^1 cannot be covered by one contractible open set. Equivalently, S^1 itself is not contractible.

However, S^1 can be covered by two contractible open arcs. For example, take two open arcs U_1, U_2 whose union is the whole circle:

$$S^1 = U_1 \cup U_2.$$

Each U_i is homeomorphic to an open interval, hence contractible.

Thus the minimal number of contractible open sets needed to cover S^1 is

$$\boxed{2}.$$

This agrees with the cup-length bound:

$$\text{cup-length}(S^1) = 1,$$

so we need at least

$$1 + 1 = 2$$

contractible open sets.

Example 3. Why S^2 needs at least two contractible open sets

The cohomology ring of S^2 is

$$H^*(S^2) \cong \mathbb{Z}[u]/(u^2), \quad |u| = 2.$$

Here

$$u \neq 0,$$

but

$$u^2 = 0.$$

The reason is that

$$u^2 \in H^4(S^2),$$

but

$$H^4(S^2) = 0.$$

Hence

$$\text{cup-length}(S^2) = 1.$$

Therefore S^2 cannot be covered by one contractible open set.

But S^2 can be covered by two contractible open sets. Let

$$U_N = S^2 \setminus \{\text{south pole}\},$$

and

$$U_S = S^2 \setminus \{\text{north pole}\}.$$

Each of these spaces is homeomorphic to $\mathbb{R}^2$, hence contractible:

$$U_N \simeq \mathbb{R}^2, \quad U_S \simeq \mathbb{R}^2.$$

Also,

$$S^2 = U_N \cup U_S.$$

Therefore the minimal number of contractible open sets needed to cover S^2 is

$$\boxed{2}.$$

8.3 Medium Examples

Example 4. A concrete diagonal computation on T^2

Let

$$T^2 = S^1 \times S^1.$$

The cohomology ring is

$$H^*(T^2) \cong \Lambda(\alpha, \beta), \quad |\alpha| = |\beta| = 1.$$

Thus

$$\alpha^2 = 0, \quad \beta^2 = 0, \quad \beta\alpha = -\alpha\beta.$$

Let

$$\omega = \alpha\beta.$$

Then

$$\omega \in H^2(T^2)$$

is the orientation class of the torus.

Now consider

$$T^2 \times T^2.$$

Let

$$\alpha_1 = \pi_1^*(\alpha), \quad \beta_1 = \pi_1^*(\beta),$$

and

$$\alpha_2 = \pi_2^*(\alpha), \quad \beta_2 = \pi_2^*(\beta).$$

The diagonal map

$$\Delta : T^2 \rightarrow T^2 \times T^2$$

satisfies

$$\Delta^*(\alpha_1) = \alpha, \quad \Delta^*(\beta_1) = \beta,$$

and

$$\Delta^*(\alpha_2) = \alpha, \quad \Delta^*(\beta_2) = \beta.$$

Now compute

$$\Delta^*((\alpha_1 + \beta_2)(\beta_1 - \alpha_2)).$$

First expand:

$$(\alpha_1 + \beta_2)(\beta_1 - \alpha_2) = \alpha_1\beta_1 - \alpha_1\alpha_2 + \beta_2\beta_1 - \beta_2\alpha_2.$$

Apply Δ^* term by term.

First,

$$\Delta^*(\alpha_1\beta_1) = \alpha\beta = \omega.$$

Second,

$$\Delta^*(\alpha_1\alpha_2) = \alpha\alpha = 0.$$

Third,

$$\Delta^*(\beta_2\beta_1) = \beta\beta = 0.$$

Fourth,

$$\Delta^*(\beta_2\alpha_2) = \beta\alpha = -\alpha\beta = -\omega.$$

Therefore

$$\Delta^*(-\beta_2\alpha_2) = -(-\omega) = \omega.$$

Combining the four terms gives

$$\Delta^*((\alpha_1 + \beta_2)(\beta_1 - \alpha_2)) = \omega - 0 + 0 + \omega.$$

Hence

$$\boxed{\Delta^*((\alpha_1 + \beta_2)(\beta_1 - \alpha_2)) = 2\omega.}$$

This example shows why signs matter. Since α and β both have degree 1, they anticommute:

$$\beta\alpha = -\alpha\beta.$$

Example 5. Why T^2 cannot be covered by two contractible open sets

The cohomology ring of the torus is

$$H^*(T^2) \cong \Lambda(\alpha, \beta).$$

The essential fact is

$$\alpha\beta \neq 0.$$

Indeed,

$$\alpha\beta = \omega,$$

where ω is the fundamental cohomology class of T^2 .

Thus

$$\text{cup-length}(T^2) = 2.$$

Suppose, for contradiction, that

$$T^2 = U \cup V,$$

where U and V are contractible open sets.

By the vanishing theorem for contractible open covers, every product of two positive-degree cohomology classes would vanish. Therefore we would get

$$\alpha\beta = 0.$$

But this contradicts

$$\alpha\beta = \omega \neq 0.$$

Hence T^2 cannot be covered by two contractible open sets.

Since T^2 can be covered by three contractible open sets, the exact minimum is

$$\boxed{3.}$$

Example 6. The three-torus T^3

Let

$$T^3 = S^1 \times S^1 \times S^1.$$

Its cohomology ring is

$$H^*(T^3) \cong \Lambda(\alpha, \beta, \gamma),$$

where

$$|\alpha| = |\beta| = |\gamma| = 1.$$

The product

$$\alpha\beta\gamma$$

lies in

$$H^3(T^3).$$

It is nonzero and represents the orientation class of T^3 :

$$\alpha\beta\gamma \neq 0.$$

Therefore

$$\text{cup-length}(T^3) = 3.$$

If T^3 can be covered by n contractible open sets, then

$$n > \text{cup-length}(T^3) = 3.$$

Thus

$$\boxed{n \geq 4.}$$

So T^3 cannot be covered by three contractible open sets.

The pattern is:

$$T^1 \text{ needs at least } 2,$$

$$T^2 \text{ needs at least } 3,$$

$$T^3 \text{ needs at least } 4,$$

and in general,

$$\boxed{T^k \text{ needs at least } k + 1 \text{ contractible open sets.}}$$

Example 7. Product of spheres $S^m \times S^n$

Let

$$X = S^m \times S^n, \quad m, n > 0.$$

Let

$$a \in H^m(S^m), \quad b \in H^n(S^n)$$

be generators.

On

$$S^m \times S^n,$$

define

$$\alpha = \pi_1^*(a) \in H^m(S^m \times S^n),$$

and

$$\beta = \pi_2^*(b) \in H^n(S^m \times S^n).$$

Their cup product is

$$\alpha\beta \in H^{m+n}(S^m \times S^n).$$

This class is nonzero. It is the orientation class of the product manifold:

$$\alpha\beta \neq 0.$$

Therefore

$$\text{cup-length}(S^m \times S^n) = 2.$$

So any cover of $S^m \times S^n$ by contractible open sets must contain at least

$$2 + 1 = 3$$

sets.

Hence

$S^m \times S^n$ cannot be covered by two contractible open sets.

For example,

$$S^2 \times S^2$$

has a nonzero product

$$u_1 u_2 \neq 0$$

in degree 4. Therefore $S^2 \times S^2$ needs at least three contractible open sets.

This example shows that the product structure of cohomology detects more than the individual cohomology groups.

8.4 Deep Examples

Example 8. Boundary map compatibility for $(D^2, S^1) \times S^1$

Consider the pair

$$(D^2, S^1).$$

The long exact sequence of the pair contains

$$H^1(D^2) \longrightarrow H^1(S^1) \xrightarrow{\delta} H^2(D^2, S^1) \longrightarrow H^2(D^2).$$

Since

$$H^1(D^2) = 0$$

and

$$H^2(D^2) = 0,$$

the connecting homomorphism

$$\delta : H^1(S^1) \rightarrow H^2(D^2, S^1)$$

is an isomorphism.

Let

$$a \in H^1(S^1)$$

be a generator. Then

$$\delta a$$

is a generator of

$$H^2(D^2, S^1).$$

Now cross with another circle S^1 . Let

$$b \in H^1(S^1)$$

be a generator.

Then

$$a \times b \in H^2(S^1 \times S^1).$$

The relative pair becomes

$$(D^2 \times S^1, S^1 \times S^1).$$

Problem 3 gives the naturality identity

$$\delta(a \times b) = (\delta a) \times b.$$

The right-hand side is a class in

$$H^3(D^2 \times S^1, S^1 \times S^1).$$

Geometrically,

$$D^2 \times S^1$$

is a solid torus, while

$$S^1 \times S^1$$

is its boundary torus.

Thus

$$(\delta a) \times b$$

is the relative fundamental cohomology class of the solid torus relative to its boundary.

So the identity

$$\boxed{\delta(a \times b) = (\delta a) \times b}$$

says that the boundary obstruction on S^1 , crossed with another S^1 , becomes the relative volume class of the solid torus.

In words:

the boundary class of the disk, crossed with the circle direction, gives the relative top class of the solid torus.

Example 9. Complex projective space $\mathbb{CP}^n$

The cohomology ring of complex projective space is

$$H^*(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}[u]/(u^{n+1}), \quad |u| = 2.$$

The class

$$u \in H^2(\mathbb{CP}^n; \mathbb{Z})$$

is called the hyperplane class.

Its powers are

$$1, u, u^2, \dots, u^n,$$

with

$$u^n \neq 0,$$

but

$$u^{n+1} = 0.$$

Therefore the cup-length is

$$\text{cup-length}(\mathbb{CP}^n) = n.$$

Indeed, the product

$$\underbrace{u \smile u \smile \cdots \smile u}_{n \text{ times}} = u^n$$

is nonzero, while the $(n+1)$ -fold product is zero.

Therefore any contractible open cover of $\mathbb{CP}^n$ must have at least

$$n+1$$

sets.

Thus

$$\mathbb{CP}^n \text{ cannot be covered by } n \text{ contractible open sets.}$$

It needs at least

$$n + 1$$

contractible open sets.

For example,

$$\mathbb{CP}^1 \cong S^2.$$

Here

$$H^*(\mathbb{CP}^1) \cong \mathbb{Z}[u]/(u^2),$$

so the cup-length is 1, and at least two contractible open sets are needed.

For $\mathbb{CP}^2$,

$$H^*(\mathbb{CP}^2) \cong \mathbb{Z}[u]/(u^3), \quad |u| = 2.$$

Here

$$u^2 \neq 0,$$

so

$$\text{cup-length}(\mathbb{CP}^2) = 2.$$

Therefore $\mathbb{CP}^2$ needs at least three contractible open sets.

This example is deeper than the torus example in one important way. For the torus, the nonzero top product comes from multiplying different degree 1 classes. For $\mathbb{CP}^n$, the obstruction comes from repeatedly multiplying one degree 2 class.

Example 10. Real projective space $\mathbb{RP}^n$

For real projective space, it is most natural to use coefficients in

$$\mathbb{F}_2.$$

The cohomology ring is

$$H^*(\mathbb{RP}^n; \mathbb{F}_2) \cong \mathbb{F}_2[t]/(t^{n+1}), \quad |t| = 1.$$

Thus

$$1, t, t^2, \dots, t^n$$

are nonzero, but

$$t^{n+1} = 0.$$

In particular,

$$t^n \neq 0.$$

Therefore

$$\text{cup-length}(\mathbb{RP}^n) = n.$$

So every contractible open cover of $\mathbb{RP}^n$ must have at least

$$n + 1$$

sets.

Hence

$$\boxed{\mathbb{RP}^n \text{ cannot be covered by } n \text{ contractible open sets.}}$$

For example, for $\mathbb{RP}^2$,

$$H^*(\mathbb{RP}^2; \mathbb{F}_2) \cong \mathbb{F}_2[t]/(t^3).$$

Here

$$t^2 \neq 0,$$

so

$$\text{cup-length}(\mathbb{RP}^2) = 2.$$

Therefore $\mathbb{RP}^2$ needs at least

$$3$$

contractible open sets.

This example is important because with ordinary $\mathbb{Z}$ -coefficients, torsion phenomena can hide some cup-product information. With $\mathbb{F}_2$ -coefficients, the powers of the degree 1 class become clearly visible.

Example 11. Wedge of circles versus the torus

Compare

$$S^1 \vee S^1$$

with

$$T^2.$$

Both spaces have two independent degree 1 cohomology classes.

For the wedge of circles,

$$H^1(S^1 \vee S^1) \cong \mathbb{Z}^2.$$

Let

$$a, b \in H^1(S^1 \vee S^1)$$

be the two generators.

However,

$$H^2(S^1 \vee S^1) = 0.$$

Therefore

$$ab = 0.$$

So the cup-length of $S^1 \vee S^1$ is

$$\text{cup-length}(S^1 \vee S^1) = 1.$$

Thus cup-length only forces at least

$$2$$

contractible open sets.

Now compare this with the torus. For T^2 ,

$$H^1(T^2) \cong \mathbb{Z}^2,$$

with generators

$$\alpha, \beta.$$

But now

$$H^2(T^2) \cong \mathbb{Z},$$

and

$$\alpha\beta \neq 0.$$

Therefore

$$\text{cup-length}(T^2) = 2.$$

So T^2 needs at least

$$3$$

contractible open sets.

The important conclusion is:

$$\boxed{H^1(S^1 \vee S^1) \cong H^1(T^2),}$$

but their cohomology rings are different.

For the wedge,

$$ab = 0,$$

while for the torus,

$$\alpha\beta \neq 0.$$

Thus the torus has a genuinely two-dimensional cup product, while the wedge of circles does not.

This example shows that cohomology groups alone are not enough. The ring structure matters.

Example 12. A space with nonzero cohomology but zero positive-degree products

Let

$$X = S^2 \vee S^2.$$

Then

$$H^2(X) \cong \mathbb{Z} \oplus \mathbb{Z}.$$

Let

$$u, v \in H^2(X)$$

be the two generators, one from each sphere.

Since X has no 4-dimensional cells,

$$H^4(X) = 0.$$

Therefore every product of degree 2 classes is zero:

$$u^2 = 0, \quad v^2 = 0, \quad uv = 0.$$

So even though

$$H^2(X) \neq 0,$$

all products of positive-degree classes vanish.

Thus

$$\text{cup-length}(S^2 \vee S^2) = 1.$$

This only gives the lower bound that at least two contractible open sets are needed.

Now compare this with

$$S^2 \times S^2.$$

There,

$$H^2(S^2 \times S^2) \cong \mathbb{Z}u_1 \oplus \mathbb{Z}u_2,$$

and

$$u_1 u_2 \neq 0.$$

The product

$$u_1 u_2$$

generates

$$H^4(S^2 \times S^2) \cong \mathbb{Z}.$$

Therefore

$$\text{cup-length}(S^2 \times S^2) = 2.$$

Hence $S^2 \times S^2$ needs at least three contractible open sets.

Again, the important difference is the product structure:

$$S^2 \vee S^2 \quad \text{has zero degree 2 products,}$$

whereas

$$S^2 \times S^2 \quad \text{has a nonzero degree 4 product.}$$

8.5 Summary Table

The following table summarizes the main examples.

Space	Cohomology ring idea	Cup-length	Cover lower bound
S^1	$\Lambda(a), a = 1$	1	≥ 2
S^2	$\mathbb{Z}[u]/(u^2), u = 2$	1	≥ 2
T^2	$\Lambda(\alpha, \beta)$	2	≥ 3
T^3	$\Lambda(\alpha, \beta, \gamma)$	3	≥ 4
T^k	$\Lambda(\alpha_1, \dots, \alpha_k)$	k	$\geq k + 1$
$S^m \times S^n$	$\alpha\beta \neq 0$	2	≥ 3
$\mathbb{CP}^n$	$\mathbb{Z}[u]/(u^{n+1})$	n	$\geq n + 1$
$\mathbb{RP}^n$	$\mathbb{F}_2[t]/(t^{n+1})$	n	$\geq n + 1$
$S^1 \vee S^1$	all positive products vanish	1	≥ 2
$S^2 \vee S^2$	all degree 2 products vanish	1	≥ 2

8.6 Conceptual Summary

The examples show three levels of understanding.

At the easy level, the examples

$$S^1 \quad \text{and} \quad S^2$$

show that a single nonzero positive-degree cohomology class already proves that a space is not contractible.

At the medium level, the examples

$$T^2, \quad T^3, \quad S^m \times S^n$$

show that nonzero products of positive-degree classes force a space to require more contractible open sets in a cover.

At the deeper level, the examples

$$\mathbb{CP}^n \quad \text{and} \quad \mathbb{RP}^n$$

show that long powers of a single class can force strong lower bounds on the size of a contractible open cover.

The central principle is:

nonzero cup products obstruct small contractible open covers.

Equivalently,

$\text{cup-length}(X) + 1 \leq \text{minimum number of contractible open sets needed to cover } X.$

Thus the richer the multiplication in the cohomology ring, the harder it is to cover the space by a small number of contractible open pieces.

9 Problem 6. Maps between closed orientable surfaces of different genus

Statement

Let Σ_g be the closed orientable surface of genus g . Suppose

$$\varphi : \Sigma_g \longrightarrow \Sigma_h$$

is a continuous map, where

$$g < h.$$

Show that the induced map

$$\varphi_* : H_2(\Sigma_g) \longrightarrow H_2(\Sigma_h)$$

is the zero map.

The hint suggests considering

$$\varphi^* : H^1(\Sigma_h) \longrightarrow H^1(\Sigma_g).$$

Geometric idea

The group

$$H_2(\Sigma_g; \mathbb{Z}) \cong \mathbb{Z}$$

is generated by the fundamental class

$$[\Sigma_g].$$

Therefore the map φ_* is multiplication by an integer d :

$$\varphi_*[\Sigma_g] = d[\Sigma_h].$$

This integer d is the degree of φ , written

$$d = \deg(\varphi).$$

Thus the problem is equivalent to proving

$$\deg(\varphi) = 0.$$

The key idea is:

If $\deg(\varphi) \neq 0$, then $\varphi^* : H^1(\Sigma_h; \mathbb{Q}) \rightarrow H^1(\Sigma_g; \mathbb{Q})$ must be injective.

But

$$\dim_{\mathbb{Q}} H^1(\Sigma_h; \mathbb{Q}) = 2h$$

and

$$\dim_{\mathbb{Q}} H^1(\Sigma_g; \mathbb{Q}) = 2g.$$

Since $g < h$, we have

$$2g < 2h.$$

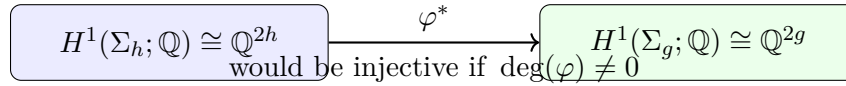
So an injective linear map

$$\mathbb{Q}^{2h} \longrightarrow \mathbb{Q}^{2g}$$

cannot exist. Therefore the degree must be zero.

Dimension diagram

The contradiction can be visualized as follows:



$2h > 2g$, so injection is impossible.

Solution

Choose orientations on Σ_g and Σ_h . Since both are closed connected orientable surfaces,

$$H_2(\Sigma_g; \mathbb{Z}) \cong \mathbb{Z}, \quad H_2(\Sigma_h; \mathbb{Z}) \cong \mathbb{Z}.$$

Let

$$[\Sigma_g] \in H_2(\Sigma_g; \mathbb{Z}), \quad [\Sigma_h] \in H_2(\Sigma_h; \mathbb{Z})$$

be the fundamental classes.

Then there exists an integer $d \in \mathbb{Z}$ such that

$$\varphi_*[\Sigma_g] = d[\Sigma_h].$$

We need to show that

$$d = 0.$$

Suppose, for contradiction, that

$$d \neq 0.$$

We work with rational coefficients. Consider the induced map

$$\varphi^* : H^1(\Sigma_h; \mathbb{Q}) \longrightarrow H^1(\Sigma_g; \mathbb{Q}).$$

We claim that φ^* is injective.

Let

$$0 \neq \alpha \in H^1(\Sigma_h; \mathbb{Q}).$$

By Poincaré duality on the closed orientable surface Σ_h , the pairing

$$H^1(\Sigma_h; \mathbb{Q}) \times H^1(\Sigma_h; \mathbb{Q}) \longrightarrow \mathbb{Q}$$

given by

$$(\alpha, \beta) \longmapsto \langle \alpha \smile \beta, [\Sigma_h] \rangle$$

is nondegenerate. Therefore there exists

$$\beta \in H^1(\Sigma_h; \mathbb{Q})$$

such that

$$\langle \alpha \smile \beta, [\Sigma_h] \rangle \neq 0.$$

Now use naturality of the cup product and the definition of degree:

$$\begin{aligned} \langle \varphi^* \alpha \smile \varphi^* \beta, [\Sigma_g] \rangle &= \langle \varphi^* (\alpha \smile \beta), [\Sigma_g] \rangle \\ &= \langle \alpha \smile \beta, \varphi_* [\Sigma_g] \rangle \\ &= \langle \alpha \smile \beta, d[\Sigma_h] \rangle \\ &= d \langle \alpha \smile \beta, [\Sigma_h] \rangle. \end{aligned}$$

Since

$$d \neq 0$$

and

$$\langle \alpha \smile \beta, [\Sigma_h] \rangle \neq 0,$$

we obtain

$$\langle \varphi^* \alpha \smile \varphi^* \beta, [\Sigma_g] \rangle \neq 0.$$

In particular,

$$\varphi^* \alpha \neq 0.$$

Thus every nonzero class in $H^1(\Sigma_h; \mathbb{Q})$ has nonzero pullback. Therefore

$$\varphi^* : H^1(\Sigma_h; \mathbb{Q}) \rightarrow H^1(\Sigma_g; \mathbb{Q})$$

is injective.

But

$$\dim_{\mathbb{Q}} H^1(\Sigma_h; \mathbb{Q}) = 2h$$

and

$$\dim_{\mathbb{Q}} H^1(\Sigma_g; \mathbb{Q}) = 2g.$$

Since $g < h$,

$$2g < 2h.$$

So an injective linear map

$$\mathbb{Q}^{2h} \longrightarrow \mathbb{Q}^{2g}$$

cannot exist.

This contradiction shows that our assumption was false. Hence

$$d = 0.$$

Therefore

$$\varphi_*[\Sigma_g] = 0.$$

Since $H_2(\Sigma_g)$ is generated by $[\Sigma_g]$, it follows that

$\varphi_* : H_2(\Sigma_g) \longrightarrow H_2(\Sigma_h)$ is the zero map.
--

Summary chart

Surface	$H^1(-; \mathbb{Q})$	$\dim_{\mathbb{Q}} H^1(-; \mathbb{Q})$
Σ_g	$\mathbb{Q}^{2g}$	$2g$
Σ_h	$\mathbb{Q}^{2h}$	$2h$

Since $g < h$, one has

$$2g < 2h.$$

Thus a nonzero-degree map would force an impossible injection

$$H^1(\Sigma_h; \mathbb{Q}) \hookrightarrow H^1(\Sigma_g; \mathbb{Q}).$$

Therefore the degree is zero.

10 Problem 7. The Hopf invariant

Statement

Given a map

$$\phi : S^{2n-1} \longrightarrow S^n,$$

let

$$X_\phi = S^n \cup_\phi D^{2n}.$$

The cohomology of X_ϕ has the form

$$H^*(X_\phi) = \langle 1, x_n, x_{2n} \rangle,$$

where

$$x_i \in H^i(X_\phi).$$

Thus

$$x_n^2 = H(\phi)x_{2n}$$

for some integer

$$H(\phi) \in \mathbb{Z}.$$

(a) Show that the map

$$[\phi] \longmapsto H(\phi)$$

defines a homomorphism

$$H : \pi_{2n-1}(S^n) \longrightarrow \mathbb{Z}.$$

The integer $H(\phi)$ is called the Hopf invariant.

(b) By considering a cell decomposition of $S^n \times S^n$, show that H is nontrivial for every even n . Deduce that

$$\pi_{4m-1}(S^{2m})$$

is infinite for all $m > 0$.

10.1 The mapping cone X_ϕ

The space

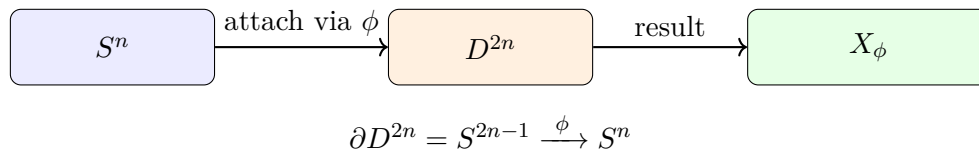
$$X_\phi = S^n \cup_\phi D^{2n}$$

is obtained by attaching a $2n$ -cell to S^n using the attaching map

$$\phi : S^{2n-1} \rightarrow S^n.$$

Thus X_ϕ has one cell in dimension 0, one cell in dimension n , and one cell in dimension $2n$.

Schematically:



The cellular cochain groups are

$$C^0(X_\phi) \cong \mathbb{Z}, \quad C^n(X_\phi) \cong \mathbb{Z}, \quad C^{2n}(X_\phi) \cong \mathbb{Z},$$

and there are no cells in the dimensions between n and $2n$. Therefore

$$H^0(X_\phi) \cong \mathbb{Z}, \quad H^n(X_\phi) \cong \mathbb{Z}, \quad H^{2n}(X_\phi) \cong \mathbb{Z}.$$

Let

$$x_n \in H^n(X_\phi), \quad x_{2n} \in H^{2n}(X_\phi)$$

be generators. Then the product

$$x_n^2 = x_n \smile x_n$$

lies in

$$H^{2n}(X_\phi) \cong \mathbb{Z}.$$

Hence there is a unique integer $H(\phi)$ such that

$$x_n^2 = H(\phi)x_{2n}.$$

This integer is the Hopf invariant of ϕ .

10.2 Part (a). The Hopf invariant is a homomorphism

Step 1. Dependence only on the homotopy class

If

$$\phi \simeq \psi : S^{2n-1} \rightarrow S^n,$$

then the corresponding cell complexes

$$X_\phi = S^n \cup_\phi D^{2n}$$

and

$$X_\psi = S^n \cup_\psi D^{2n}$$

are homotopy equivalent. The equivalence preserves the n -cell and the $2n$ -cell up to orientation choices. Therefore the coefficient of x_{2n} in the square x_n^2 is unchanged up to the corresponding orientation convention.

Thus

$$H(\phi) = H(\psi).$$

So the Hopf invariant depends only on the homotopy class

$$[\phi] \in \pi_{2n-1}(S^n).$$

Step 2. Odd-dimensional case

If n is odd, then graded commutativity gives

$$x_n^2 = (-1)^{n^2} x_n^2 = (-1)^n x_n^2 = -x_n^2.$$

Hence

$$2x_n^2 = 0.$$

But

$$H^{2n}(X_\phi) \cong \mathbb{Z}$$

has no torsion. Therefore

$$x_n^2 = 0.$$

Thus

$$H(\phi) = 0$$

for every ϕ when n is odd.

So in odd dimensions, the Hopf invariant is the zero homomorphism:

$$H : \pi_{2n-1}(S^n) \rightarrow \mathbb{Z} \text{ is identically zero if } n \text{ is odd.}$$

Step 3. Additivity

Now suppose n is even.

Let

$$\phi, \psi : S^{2n-1} \rightarrow S^n$$

represent elements of

$$\pi_{2n-1}(S^n).$$

Their sum

$$\phi + \psi$$

is represented by first pinching the domain sphere to a wedge and then applying ϕ on one summand and ψ on the other:

$$S^{2n-1} \xrightarrow{\text{pinch}} S^{2n-1} \vee S^{2n-1} \xrightarrow{\phi \vee \psi} S^n.$$

Diagrammatically:

$$S^{2n-1} \xrightarrow{\text{pinch}} S^{2n-1} \vee S^{2n-1} \xrightarrow{\phi \vee \psi} S^n$$

The Hopf invariant can also be described cochain-theoretically.

Let

$$u \in C^n(S^n)$$

be a cocycle representing the generator of

$$H^n(S^n; \mathbb{Z}).$$

Since

$$H^n(S^{2n-1}; \mathbb{Z}) = 0,$$

the pullback

$$\phi^*u$$

is exact. Hence there exists a cochain

$$v_\phi \in C^{n-1}(S^{2n-1})$$

such that

$$dv_\phi = \phi^*u.$$

One cochain formula for the Hopf invariant is

$$H(\phi) = \langle v_\phi \smile \phi^*u, [S^{2n-1}] \rangle.$$

Equivalently,

$$H(\phi) = \langle v_\phi \smile dv_\phi, [S^{2n-1}] \rangle.$$

For the sum $\phi + \psi$, the domain sphere may be viewed as two large pieces joined along a thin region. On one piece the map is ϕ , and on the other piece the map is ψ . A primitive for $(\phi + \psi)^*u$ can be chosen to agree with v_ϕ on the first piece and with v_ψ on the second piece.

Since the two pieces contribute independently, the integral splits:

$$H(\phi + \psi) = H(\phi) + H(\psi).$$

Therefore

$$\boxed{H([\phi] + [\psi]) = H([\phi]) + H([\psi]).}$$

So

$$\boxed{H : \pi_{2n-1}(S^n) \longrightarrow \mathbb{Z}}$$

is a group homomorphism.

10.3 Part (b). Nontriviality for even n

We now show that for every even n , there exists a map

$$\phi : S^{2n-1} \rightarrow S^n$$

with

$$H(\phi) \neq 0.$$

The map will arise from the top cell of

$$S^n \times S^n.$$

Cell decomposition of $S^n \times S^n$

The sphere S^n has one 0-cell and one n -cell:

$$S^n = e^0 \cup e^n.$$

Therefore

$$S^n \times S^n$$

has cells in dimensions

$$0, \quad n, \quad n, \quad 2n.$$

More precisely,

$$S^n \times S^n = e^0 \cup e_1^n \cup e_2^n \cup e^{2n}.$$

The n -skeleton is

$$S^n \vee S^n.$$

The top cell is attached by a map

$$S^{2n-1} \longrightarrow S^n \vee S^n.$$

This attaching map is the Whitehead product

$$[\iota_1, \iota_2],$$

where

$$\iota_1 : S^n \rightarrow S^n \vee S^n$$

and

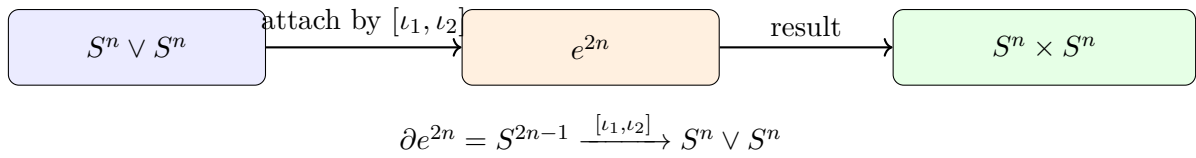
$$\iota_2 : S^n \rightarrow S^n \vee S^n$$

are the inclusions of the two wedge summands.

Thus

$$S^n \times S^n \simeq (S^n \vee S^n) \cup_{[\iota_1, \iota_2]} e^{2n}.$$

Cell-structure diagram



Now let

$$\nabla : S^n \vee S^n \longrightarrow S^n$$

be the fold map. It is defined by taking each wedge summand identically onto S^n :

$$\nabla|_{S_1^n} = \text{id}_{S^n}, \quad \nabla|_{S_2^n} = \text{id}_{S^n}.$$

Compose the attaching map with the fold map:

$$S^{2n-1} \xrightarrow{[\iota_1, \iota_2]} S^n \vee S^n \xrightarrow{\nabla} S^n.$$

Define

$$\phi = \nabla \circ [\iota_1, \iota_2].$$

This map is the Whitehead square

$$\phi = [\iota, \iota] \in \pi_{2n-1}(S^n).$$

A map from $S^n \times S^n$ to X_ϕ

Since the top cell of $S^n \times S^n$ is attached by

$$[\iota_1, \iota_2],$$

and since X_ϕ is obtained by attaching a $2n$ -cell to S^n by

$$\phi = \nabla \circ [\iota_1, \iota_2],$$

the fold map extends to a cellular map

$$q : S^n \times S^n \longrightarrow X_\phi.$$

The diagram is:

$$\begin{array}{ccc} S^{2n-1} & \xrightarrow{[\iota_1, \iota_2]} & S^n \vee S^n \\ \parallel & & \downarrow \nabla \\ S^{2n-1} & \xrightarrow{\phi} & S^n \end{array}$$

After attaching the $2n$ -cells, this gives

$$q : S^n \times S^n \rightarrow X_\phi.$$

Cohomology computation

Let

$$a = \pi_1^*(u) \in H^n(S^n \times S^n),$$

and

$$b = \pi_2^*(u) \in H^n(S^n \times S^n),$$

where

$$u \in H^n(S^n)$$

is a generator.

Then

$$H^*(S^n \times S^n)$$

has generators

$$1, \quad a, \quad b, \quad ab.$$

Here

$$ab \in H^{2n}(S^n \times S^n)$$

is the top cohomology class.

Since a and b come from the two sphere factors, we have

$$a^2 = 0, \quad b^2 = 0.$$

Also,

$$ba = (-1)^n ab.$$

Now let

$$x = x_n \in H^n(X_\phi), \quad y = x_{2n} \in H^{2n}(X_\phi).$$

By construction, the fold map sends the n -dimensional generator of X_ϕ to the sum of the two generators in $S^n \times S^n$. Therefore

$$q^*(x) = a + b.$$

Also, choosing orientations consistently,

$$q^*(y) = ab.$$

Now compute:

$$q^*(x^2) = q^*(x)^2.$$

Since

$$q^*(x) = a + b,$$

we get

$$q^*(x^2) = (a + b)^2.$$

Expand:

$$(a + b)^2 = a^2 + ab + ba + b^2.$$

Using

$$a^2 = 0, \quad b^2 = 0, \quad ba = (-1)^n ab,$$

we obtain

$$(a + b)^2 = ab + (-1)^n ab.$$

Hence

$$(a + b)^2 = (1 + (-1)^n)ab.$$

On the other hand, by definition of the Hopf invariant,

$$x^2 = H(\phi)y.$$

Applying q^* , we get

$$q^*(x^2) = H(\phi)q^*(y) = H(\phi)ab.$$

Therefore

$$H(\phi)ab = (1 + (-1)^n)ab.$$

Since $ab \neq 0$, it follows that

$$\boxed{H(\phi) = 1 + (-1)^n.}$$

Thus

$$H(\phi) = \begin{cases} 2, & n \text{ even,} \\ 0, & n \text{ odd.} \end{cases}$$

In particular, when n is even,

$$\boxed{H(\phi) = 2 \neq 0.}$$

Therefore H is nontrivial for every even n .

Summary chart for $S^n \times S^n$

Object	Description	Role
$S^n \vee S^n$	n -skeleton of $S^n \times S^n$	two n -cells
e^{2n}	top cell	creates H^{2n}
$[\iota_1, \iota_2]$	$S^{2n-1} \rightarrow S^n \vee S^n$	attaching map of top cell
∇	$S^n \vee S^n \rightarrow S^n$	fold map
$\phi = \nabla \circ [\iota_1, \iota_2]$	$S^{2n-1} \rightarrow S^n$	Whitehead square
$H(\phi)$	Hopf invariant	$1 + (-1)^n$

10.4 Deduction: $\pi_{4m-1}(S^{2m})$ is infinite

Now take

$$n = 2m.$$

Then n is even. From the previous subsection, there exists

$$\phi : S^{2n-1} \rightarrow S^n$$

such that

$$H(\phi) = 2 \neq 0.$$

Therefore the homomorphism

$$H : \pi_{2n-1}(S^n) \rightarrow \mathbb{Z}$$

has nonzero image.

A nonzero subgroup of $\mathbb{Z}$ is infinite. Hence

$$\pi_{2n-1}(S^n)$$

must be infinite.

Since

$$n = 2m,$$

we have

$$2n - 1 = 4m - 1.$$

Therefore

$$\pi_{2n-1}(S^n) = \pi_{4m-1}(S^{2m}).$$

Thus

$$\pi_{4m-1}(S^{2m}) \text{ is infinite for every } m > 0.$$

10.5 Conceptual summary

Problem 6 uses the interaction between degree, cup products, and Poincaré duality.

If

$$\deg(\varphi) \neq 0,$$

then the pullback

$$\varphi^* : H^1(\Sigma_h; \mathbb{Q}) \rightarrow H^1(\Sigma_g; \mathbb{Q})$$

would preserve enough of the intersection pairing to be injective. But this is impossible when $g < h$, because

$$\dim H^1(\Sigma_h; \mathbb{Q}) = 2h > 2g = \dim H^1(\Sigma_g; \mathbb{Q}).$$

Therefore

$$\varphi_* : H_2(\Sigma_g) \rightarrow H_2(\Sigma_h)$$

is zero.

Problem 7 uses cup products in a two-cell complex

$$X_\phi = S^n \cup_\phi D^{2n}.$$

The Hopf invariant $H(\phi)$ measures the square of the middle-dimensional cohomology class:

$$x_n^2 = H(\phi)x_{2n}.$$

The cell decomposition

$$S^n \times S^n = (S^n \vee S^n) \cup_{[\iota_1, \iota_2]} e^{2n}$$

shows that the Whitehead square

$$[\iota, \iota] \in \pi_{2n-1}(S^n)$$

has Hopf invariant

$$H([\iota, \iota]) = 1 + (-1)^n.$$

Hence for even n ,

$$H([\iota, \iota]) = 2 \neq 0.$$

Therefore the Hopf invariant homomorphism is nontrivial for even n , and consequently

$$\pi_{4m-1}(S^{2m})$$

is infinite for all $m > 0$.